

When “Milliseconds” Matter: The Real Effects of High-Frequency Trading*

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Abstract

This paper examines how high-frequency trading affects corporate investment and financing. Using a unique Australian setting with broker-level HFT identifiers, we exploit the introduction of ASX-ITCH, a high-speed order-level market data feed that increased the speed of trading information, as plausibly exogenous variation in HFT presence. As HFT activity rises, firms experience a sizable decline in investment and equity issuance. These effects are concentrated among firms that rely more heavily on external equity financing. We find corroborating evidence in the staggered global roll-out of exchange co-location services. Overall, the findings show that financial market innovations can alter firms’ financing capacity and investment decisions, providing a channel through which financial market structure may influence aggregate economic growth.

Keywords: Financial market innovation; High-frequency trading; Real firm decisions; External financing; Economic growth

JEL Codes: G23; G31; G32; O4

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1 Introduction

High-frequency trading (HFT), one of the most consequential financial market innovations of the past two decades, now accounts for a substantial share of trading volume in modern equity markets.¹ By design, HFT combines fast market data, automated decision rules, and low-latency execution to trade at horizons measured in milliseconds. A large literature studies how HFT affects market quality, but the empirical evidence remains mixed. Many studies document improvements in liquidity and short-horizon price discovery (Hendershott, Jones, and Menkveld, 2011; Brogaard, Hendershott, and Riordan, 2014). Others show that faster trading can reduce incentives to acquire fundamental information (Weller, 2018) or amplify commonality and return comovement (Malceniece, Malcenieks, and Putniņš, 2019). Despite this active debate, comparatively little is known about whether HFT affects firms' real decisions—and, ultimately, the real economy. In this paper, we fill this gap by providing evidence on how HFT affects firms' financing and investment activities.

Conceptually, the sign of HFT's real effects is ambiguous. If HFT improves liquidity and lowers transaction costs, it can reduce firms' cost of external financing and, in turn, raise investment. Hendershott, Jones, and Menkveld (2011) provides evidence consistent with this mechanism by showing that algorithmic trading improves liquidity. However, the relation between speed and market quality need not be monotonic. When HFT increases return comovement and liquidity commonality (Malceniece, Malcenieks, and Putniņš, 2019), it can raise systematic risk exposures and potentially increase the cost of capital. Moreover, HFT liquidity provision can be fragile and may withdraw during periods of market stress, worsening effective liquidity and potentially increasing the cost of equity capital (Kirilenko, Kyle, Samadi, and Tuzun, 2017). By shaping the informational content of prices, HFT can also influence investment through feedback and learning channels emphasized in the

¹For example, using message-level data from NASDAQ, Carrion (2013) identifies HFT participation directly. Using industry estimates, the Congressional Research Service reports that HFT accounted for as much as about two-thirds of U.S. equity trades between 2008 and 2011 and that this share declined to about one-half by 2012 (Miller and Shorter, 2014).

broader literature on the real effects of financial markets (Bond, Edmans, and Goldstein, 2012). More informative prices can guide managerial learning and improve the allocation of capital (Dow and Gorton, 1997; Chen, Goldstein, and Jiang, 2007), potentially reducing misinvestment. Conversely, if HFT crowds out information acquisition in the longer term (Weller, 2018), it may weaken price signals and reduce investment efficiency. These competing mechanisms underscore the importance of clean empirical evidence on whether, and through which channels, HFT affects firms’ real decisions.

Although there are early attempts to study the real effects of HFT and algorithmic trading, the evidence remains limited and the small set of existing findings is not fully consistent with one another (Kim, Oh, Seo, and Zuo, 2026; Aliyev, Huseynov, and Rzayev, 2025). Two related challenges make it difficult to estimate the causal effect of HFT on firm decisions. First, fundamentals and market conditions jointly determine HFT activity and corporate investment. Second, researchers rarely observe a direct and clean measure of HFT participation. Instead, much of the literature relies on indirect proxies such as message-to-trade ratios, cancellations, or trade size, which can be noisy and difficult to interpret. Our setting overcomes these limitations. We use Australian data from the Securities Industry Research Centre of Asia-Pacific (SIRCA) that include broker identifiers. SIRCA classifies these identifiers into trader categories, allowing us to directly measure HFT order and trade activity. We further exploit an exchange technology upgrade—the introduction of ASX-ITCH—as plausibly exogenous variation in the feasibility and intensity of HFT.²

The ASX rolled out ASX-ITCH on April 2, 2012. This ultra-low-latency protocol distributes market data to any participant willing to pay a monthly subscription fee. The ASX describes ITCH as a premium, ultra-low-latency protocol for accessing ASX price data.³ The upgrade primarily relaxes technological constraints on data access. The fastest strategies

²Goldstein, Kwan, and Philip (2023) use the ASX-ITCH setting to study high-frequency trading strategies and market quality.

³See the ASX page “How to access ASX price data,” which states that users can connect via the “premium ultra-low latency ASX ITCH protocol”: <https://www.asx.com.au/connectivity-and-data/information-services/price-data/how-to-access-asx-price-data>.

have the highest willingness to pay for incremental latency reductions. The introduction of ITCH therefore provides a plausibly exogenous shock that should disproportionately benefit HFT. Goldstein, Kwan, and Philip (2023) show that HFT response times improve substantially after ITCH adoption, whereas the response times of slower participants remain largely unchanged. Therefore, the rollout of ASX-ITCH provides a natural setting to estimate the causal effects of HFT on firms’ real decisions.

To estimate the effect of HFT on firm investment, we first implement a difference-in-differences (DiD) design around the introduction of ASX-ITCH. The DiD compares firms with different levels of pre-ITCH exposure to HFT before versus after the ITCH rollout. Specifically, we measure this *ex-ante* exposure as the share of HFT-initiated trading volume in a firm’s stock. We scale this volume by total trading volume and average the measure over 2010–2011. We assign firms in the top tercile of this exposure measure to the treated group and firms in the bottom tercile to the control group. The baseline DiD includes firm fixed effects, year fixed effects, and SIC2 $\times$ Size Group fixed effects. Firm fixed effects absorb time-invariant firm heterogeneity, while year fixed effects absorb shocks common to all firms. The SIC2 $\times$ Size Group fixed effects further absorb unobservable differences jointly related to industry and firm size. This fixed-effects structure substantially reduces the scope for persistent firm, size, or industry characteristics to drive the estimates.

The baseline DiD estimates show that ASX-ITCH reduces corporate investment. Capex falls by 1.486 percentage points and Δ PPE falls by 2.870 percentage points for treated firms relative to control firms. The IPW and continuous-exposure specifications produce the same conclusion.

Although the fixed effects absorb several important sources of unobservable heterogeneity, low-exposure firms may still differ from high-exposure firms in pre-event firm characteristics. To improve the comparability of the two groups, we combine DiD with inverse-probability weighting (IPW). We estimate treatment probabilities from pre-event SIC2 industry, lagged log market capitalization, and lagged log trading volume. We select these variables because

they should be closely related to HFT activity. Firm size and trading volume capture the scale of trading opportunities and the potential gains from speed, while SIC2 indicators capture systematic differences in trading environments across sectors. We average the resulting weights within each firm during the pre-treatment period and apply the firm-level weight to the full panel. Following the weighted DiD approach in [Abadie \(2005\)](#) and [Acemoglu, Naidu, Restrepo, and Robinson \(2019\)](#), IPW reweights the sample so that treated and control firms are more comparable in their pre-treatment firm characteristics. It therefore reduces differences in treatment assignment related to these characteristics while preserving the DiD adjustment for time-invariant firm heterogeneity. The IPW estimates produce the same qualitative investment conclusions as the baseline DiD. We also estimate a continuous DiD by interacting the continuous HFT exposure measure with a post-ITCH indicator. This specification does not rely on a discrete treated/control assignment and yields the same qualitative investment conclusions. We also use dynamic DiD estimates and randomized placebo tests to address remaining concerns about residual selection and omitted heterogeneity.

We next examine external financing. Equity issuance falls by 6.835 percentage points, whereas debt issuance is statistically unchanged. This equity-specific response suggests limited substitution toward, or disruption of, debt financing. We further document that the declines in both investment and equity issuance concentrate among firms that relied more heavily on equity financing before ITCH. This pattern suggests that the shock affects corporate outcomes primarily through equity markets rather than through broad balance-sheet constraints. The joint response of equity issuance and investment is consistent with a financing-based mechanism. A deterioration in equity market conditions reduces firms' ability or willingness to raise external equity and, in turn, curtails real investment (e.g., [Bond, Edmans, and Goldstein 2012](#); [Grullon, Michenaud, and Weston 2015](#)). Consistent with this interpretation, both implied cost-of-equity measures rise significantly following ASX-ITCH. The baseline estimates imply increases of 4.2 and 5.4 percentage points in firms' annual required return on equity, or approximately 420 and 540 basis points. The corresponding

IPW estimates imply increases of 4.4 and 6.1 percentage points. Economically, these changes raise the hurdle rate applied to prospective projects. Fewer investments can generate returns sufficient to compensate equity investors.

To distinguish why the required return on equity rises, we examine return comovement, price efficiency, and liquidity. We measure return comovement as the logit transformation of the market-model R^2 . It therefore captures the relative importance of market-wide versus firm-specific return variation. The baseline estimate implies that the ratio of market-explained to firm-specific variation rises by approximately 35%. The IPW estimate implies an increase of approximately 32%. Thus, treated stocks' returns become materially more exposed to common market movements after ITCH. This pattern indicates greater exposure to systematic risk. Untabulated variance decompositions also show a significant decline in idiosyncratic return variance after ITCH. In contrast, the coefficients for price delay and bid-ask spreads are statistically insignificant in both specifications. A deterioration in average spread liquidity is therefore unlikely to explain the higher cost of equity. We also find no evidence that ITCH changes the speed with which information enters prices. Instead, the economically meaningful increase in the common component of returns is consistent with a risk-based explanation for the higher required returns documented in our implied cost-of-equity tests.

Overall, we find no evidence that HFT improves investment- q sensitivity; in the co-location setting, the interaction is negative and statistically significant for Capex and statistically insignificant for Δ PPE. This pattern indicates that HFT primarily affects the level of investment and external equity financing rather than improving the allocation of investment across firms. ROA is statistically unchanged, while sales growth declines across the baseline, IPW, and continuous-exposure specifications. This evidence further weighs against an interpretation based on improved resource allocation. Dynamic DiD estimates provide no evidence of differential pre-trends. Randomized treatment-assignment placebo tests place the actual investment, equity-issuance, and comovement estimates in the tails of their respective

placebo distributions.

Taken together, our findings are consistent with the introduction of ASX-ITCH worsening equity financing conditions. Firms appear to respond primarily by scaling back the level of fixed investment rather than by reallocating investment toward more efficient projects. In other words, our evidence suggests that a central consequence of HFT in our setting is that firms undertake fewer real projects, particularly when they historically depend on equity issuance.

We also examine the staggered rollout of exchange co-location services. Prior work uses these rollouts to study algorithmic and high-frequency trading across markets (Boehmer, Fong, and Wu, 2021; Aitken, Cumming, and Zhan, 2015, 2023).⁴ Co-location lets traders place their servers near an exchange’s matching engine. This proximity reduces latency and makes speed more valuable. Similar investment, equity-financing, and financing-cost patterns emerge around co-location. Exchanges choose when to adopt co-location, however, and adoption may coincide with other market changes. We therefore use this setting as corroborating evidence that the ASX-ITCH findings do not depend entirely on one Australian technology change. The ASX-ITCH setting remains our main identification design.

Collectively, our evidence shows that financial market innovations that favor speed can have meaningful real consequences, including lower corporate investment and reduced access to external equity financing. These results have practical implications for regulators and exchanges. These institutions evaluate market design choices that affect the arms race for speed and assess the benefits and costs of HFT in an ongoing policy debate.

Our paper contributes to two strands of the literature. First, we contribute to the broader literature on the real effects of financial markets. We show that changes in trading technology and the microstructure of capital markets can affect firms’ cost of capital and real investment (e.g., Bond, Edmans, and Goldstein 2012; Grullon, Michenaud, and Weston 2015; Deng,

⁴Co-location dates in the literature sometimes conflict (Aitken, Cumming, and Zhan, 2023). We reconcile the dates in Boehmer, Fong, and Wu (2021) and Aitken, Cumming, and Zhan (2023), verify the retained events using exchange press releases and contemporaneous news reports, and exclude events with unresolved dating ambiguity.

Gupta, Lipson, and Mortal 2023; Aquilina, Ibikunle, Rzayev, and Wang 2025). Our evidence complements recent work studying how algorithmic trading shapes managerial learning and other real outcomes (Kim, Oh, Seo, and Zuo, 2026; Aliyev, Huseynov, and Rzayev, 2025). It also highlights that market design choices that intensify the speed race can have first-order real effects.

Second, we contribute to the high-frequency trading and market microstructure literature by linking HFT and speed-related trading technology to real corporate outcomes. A large literature studies how HFT affects liquidity and price discovery (e.g., Carrion 2013; Hendershott, Jones, and Menkveld 2011; Brogaard, Hendershott, and Riordan 2014), information acquisition (e.g., Weller 2018), and market stability (e.g., Kirilenko, Kyle, Samadi, and Tuzun 2017). However, this literature leaves unresolved whether the controversial effects of speed on market quality translate into meaningful changes in firms’ financing and investment. To the best of our knowledge, our setting provides arguably one of the cleanest causal designs for studying how HFT affects firms’ real investment. We also propose and test a concrete channel. HFT worsens equity financing conditions by increasing firms’ cost of equity and reducing external equity issuance. The real effects concentrate among equity-issuance-reliant firms.

The rest of the paper is organized as follows. [Section 2](#) presents a simple model for our empirical study. [Section 3](#) describes the data and sample construction. [Section 4](#) presents the identification strategy and baseline results. [Section 5](#) presents the financing mechanism. [Section 6](#) evaluates the internal validity of the ASX-ITCH setting. [Section 7](#) examines what drives the increase in the cost of equity. [Section 8](#) presents additional analyses and robustness checks, including the alternative co-location setting. [Section 9](#) concludes.

2 Model

This section presents a reduced model with two blocks. The first block integrates three channels through which high-frequency trading can affect firms' cost of equity: (i) the *liquidity level channel*; (ii) the *return comovement channel*; and (iii) the *liquidity comovement channel*. A body of theoretical and empirical studies motivates this model block.⁵ The second block focuses on corporate investment decisions under financing constraints in which high-frequency trading affects equity issuance. The model provides a theoretical framework that helps interpret our empirical results.

2.1 Competing effects of HFT on cost of equity

For simplicity, we assume that the overall market can be in one of two states, denoted by $M_t = \{0, 1\}$. The value $M_t = 0$ represents a normal market. The value $M_t = 1$ represents a stressed market with high volatility, illiquidity, and abnormal pricing. Assume the probability of the stressed state is

$$\text{Prob}(M_t = 1) = p, \quad 0 < p < 1.$$

Let $\phi_i \geq 0$ denote HFT intensity in stock i , which increases for treated stocks after regulatory changes (such as ITCH). We assume a reduced-form price impact function:

$$\kappa_i(M_t, \phi_i) = \kappa_{i0} - a_i \phi_i + b_i \phi_i M_t, \quad a_i > 0, \quad b_i > 0. \quad (1)$$

The term $-a_i \phi_i$ captures the normal-state component of HFT's effect on price impact, without implying an empirical decline in bid-ask spreads. The term $b_i \phi_i M_t$ captures the fragile liquidity effect when the market becomes stressed, as HFT liquidity provision becomes more state-dependent, order book depth is withdrawn more quickly, and price impact rises. Sup-

⁵In particular, the model is consistent with the empirical work of Malceni e, Malceni eks, and Putni s (2019). We can also extend it to accommodate other competing channels, such as the *monitoring and price-informativeness channel* in Rosu, Sojli, and Tham (2021). More quoting and faster monitoring of market makers can lower expected returns and the cost of capital.

pose the second market-stress effect dominates

$$b_i > a_i > 0. \quad (2)$$

Then the average liquidity-level effect of HFT is

$$\frac{\partial}{\partial \phi_i} \mathbb{E}[\kappa_i(M_t, \phi_i)] = pb_i - a_i,$$

which has an ambiguous sign determined by the magnitude of p . If $p < a_i/b_i$, then the average liquidity-level effect is negative; otherwise, it is non-negative. This means the empirical bid–ask spread may not necessarily decline when HFT intensity rises.

However, liquidity does deteriorate in the bad state (i.e., conditional on $M_t = 1$):

$$\frac{\partial}{\partial \phi_i} \kappa_i(M_t = 1, \phi_i) = b_i - a_i > 0.$$

Therefore, the reduced-form model combines an ambiguous liquidity-level effect with liquidity deterioration in stressed states. A microfoundation of equation (1) can be established by extending the strategic trading model of [Chen, Hu, and Yang \(2025\)](#), which suggests high-frequency market makers may switch from providing liquidity to demanding liquidity when their inventory levels hit certain boundaries in stressed states.

Return comovement channel. In bad states, systematic rebalancing, ETF/index flows, deleveraging, and/or risk-off trading can generate correlated sell pressure. We assume the net liquidity supply pressure is

$$y_{it} = q_i M_t + u_{it}, \quad q_i > 0, \quad (3)$$

where q_i is the sell pressure experienced by stock i when the market becomes stressed and u_{it} is the idiosyncratic noise. Then y_{it} reflects the net sell pressure that must be absorbed

by liquidity suppliers. The stock return contains both the price-pressure component and an idiosyncratic shock:

$$r_{it} = -\kappa_i(M_t, \phi_i)y_{it} + \varepsilon_{it} = -\kappa_i(M_t, \phi_i)(q_i M_t + u_{it}) + \varepsilon_{it},$$

Market returns are lower in bad states. Suppose the market return is of the form:

$$r_t^M = -mM_t + \eta_t, \quad m > 0, \quad (4)$$

where $\text{Var}(\eta_t) = \sigma_\eta^2$. If those idiosyncratic terms satisfy

$$\mathbb{E}[u_{it} \mid M_t] = 0, \quad \mathbb{E}[\varepsilon_{it} \mid M_t] = 0, \quad \text{Cov}(u_{it}, \eta_t \mid M_t) = 0, \quad \text{Cov}(\varepsilon_{it}, \eta_t) = 0, \quad \mathbb{E}[\eta_t \mid M_t] = 0,$$

then the variance of market return is $\text{Var}(r_t^M) = m^2 p(1-p) + \sigma_\eta^2$, and

$$\text{Cov}(r_{it}, M_t) = p(1-p) (\mathbb{E}[r_{it} \mid M_t = 1] - \mathbb{E}[r_{it} \mid M_t = 0]) = -p(1-p)q_i\kappa_i(1, \phi_i).$$

Since $r_t^M = -mM_t + \eta_t$ and the idiosyncratic stock return shocks are orthogonal to η_t , the stock beta is

$$\beta_i^R(\phi_i) = \frac{\text{Cov}(r_{it}, r_t^M)}{\text{Var}(r_t^M)} = \frac{-m \text{Cov}(r_{it}, M_t)}{\text{Var}(r_t^M)} = \frac{mp(1-p)q_i}{m^2 p(1-p) + \sigma_\eta^2} [\kappa_{i0} + (b_i - a_i)\phi_i].$$

The comparative static is

$$\frac{\partial \beta_i^R(\phi_i)}{\partial \phi_i} = \frac{mp(1-p)q_i(b_i - a_i)}{m^2 p(1-p) + \sigma_\eta^2} > 0 \quad \text{iff } b_i > a_i, \quad m > 0, \quad q_i > 0.$$

Proposition 1 (HFT increases stock return beta) *If HFT raises bad-state price impact more than it reduces normal-state price impact, then HFT increases the beta of stock returns.*

Proposition 1 maps HFT into greater exposure to common market risk: higher HFT intensity ϕ_i makes the stock's return more sensitive to the common bad state M_t . This greater common-factor sensitivity is consistent with higher return comovement.

Liquidity comovement channel. The market-wide illiquidity can be captured by taking

$$\kappa_t^M = \kappa_0^M + \kappa_1^M M_t, \quad \kappa_1^M > 0. \quad (5)$$

In our two-state model, the stock's liquidity beta can be defined and computed as

$$\beta_i^L(\phi_i) = \frac{\text{Cov}(\kappa_i(M_t, \phi_i), \kappa_t^M)}{\text{Var}(\kappa_t^M)} = \frac{\kappa_i(1, \phi_i) - \kappa_i(0, \phi_i)}{\kappa_1^M} = \frac{b_i \phi_i}{\kappa_1^M}. \quad (6)$$

Therefore

$$\frac{\partial \beta_i^L(\phi_i)}{\partial \phi_i} = \frac{b_i}{\kappa_1^M} > 0. \quad (7)$$

Proposition 2 (HFT increases liquidity comovement) *If HFT increases the bad-state price impact ($b_i > 0$), then HFT also increases the stock's liquidity beta.*

Proposition 2 captures the liquidity-commonality channel: when HFT makes stock-level liquidity more sensitive to market-wide liquidity stress, investors require compensation for holding stocks that become illiquid when aggregate liquidity is stressed.

Cost of equity with competing HFT channels. The stock's required excess return can be written as

$$r_i^E(\phi_i) - r_f = \lambda_a \mathbb{E}[\kappa_i(M_t, \phi_i)] + \lambda_R \beta_i^R(\phi_i) + \lambda_L \beta_i^L(\phi_i), \quad (8)$$

where $\lambda_a > 0$ is the price of average market illiquidity, $\lambda_R > 0$ is the premium of return systematic risk, and $\lambda_L > 0$ is the premium of liquidity risk. The three terms in (8) represent the three channels through which HFT affects stock expected returns: (i) the

average liquidity-level channel; (ii) the return-comovement channel; and (iii) the liquidity-comovement channel. Differentiating (8) with respect to ϕ_i gives

$$\frac{\partial r_i^E}{\partial \phi_i} = \lambda_a(pb_i - a_i) + \lambda_R \frac{q'_i}{m}(b_i - a_i) + \lambda_L \frac{b_i}{\kappa_1^M},$$

where $q'_i = \frac{m^2 p(1-p)q_i}{m^2 p(1-p) + \sigma_\eta^2}$ satisfies $0 < q'_i \leq q_i$. Given $b_i > a_i > 0$, the return-comovement and liquidity-comovement terms are both positive, while the average liquidity-level term has an ambiguous sign depending on whether p is greater or smaller than a_i/b_i .

Proposition 3 (The net effect of HFT on the cost of equity) *The net effect of HFT on the expected stock return or the cost of equity can be decomposed as*

$$\frac{\partial r_i^E}{\partial \phi_i} = \underbrace{\lambda_a(pb_i - a_i)}_{\text{liquidity-level effect}} + \underbrace{\lambda_R \frac{q'_i}{m}(b_i - a_i)}_{\text{return comovement cost}} + \underbrace{\lambda_L \frac{b_i}{\kappa_1^M}}_{\text{liquidity comovement cost}}. \quad (9)$$

HFT can raise the cost of equity ($\frac{\partial r_i^E}{\partial \phi_i} > 0$) as long as the cost of return- and liquidity-comovement risks dominates:

$$\lambda_R \frac{q'_i}{m}(b_i - a_i) + \lambda_L \frac{b_i}{\kappa_1^M} > \lambda_a(a_i - pb_i). \quad (10)$$

While market fragmentation is a plausible mechanism in fragmented markets (because HFT links venues and can synchronize trading across them), related evidence suggests that market fragmentation is not necessary for HFT to affect systematic risk and financing costs.⁶ Our mechanism does not require fragmentation, and the key prediction is that greater HFT exposure increases the sensitivity of liquidity and returns to a common market state. A shared funding or risk-bearing constraint provides one microfoundation for this state dependence, while correlated algorithmic strategies or systematic arbitrage can reinforce it.

⁶For example, Aquilina, Ibikunle, Rzayev, and Wang (2025) replicate their HFT analysis in Hong Kong, which they characterize as an unfragmented market, and obtain qualitatively similar results. They interpret this evidence as showing that their findings are not driven by market fragmentation.

In our setting, fragmentation can explain the results only if the ITCH shock materially changed firms' exposure to fragmentation. We therefore distinguish these mechanisms empirically by examining information delay, liquidity commonality, and the state dependence of liquidity. In particular, greater liquidity comovement and stronger liquidity deterioration during aggregate stress, without improved price delay, are more consistent with common liquidity/risk-bearing exposure than with a pure information-diffusion mechanism.

2.2 Corporate investments with equity financing

We now connect the HFT-induced increase in cost of equity to firms' equity issuance and investment decisions. The objective of this model block is to capture a financing capacity mechanism rather than a productivity shock. Specifically, HFT affects the market price at which firms can sell equity claims.

Suppose firm i 's operating payoff at $t = 1$ from its investment I_i at $t = 0$ is

$$V_i(I_i) = A_i + \theta_i I_i - \frac{g_i}{2} I_i^2, \quad \theta_i > 0, \quad g_i > 0. \quad (11)$$

Suppose the firm has internal funds W_i and debt capacity $\bar{D}_i$ so that its total non-equity financing capacity is $W_i + \bar{D}_i \geq 0$. Consistent with the *pecking order theory*, any investment beyond $W_i + \bar{D}_i$ must be financed by external equity. Thus, the dollar amount of external equity financing required is $E_i = (I_i - W_i - \bar{D}_i)^+$. External equity is costly because of issuance discounts, underpricing, underwriting fees, placement frictions, and liquidity-risk compensation. Let $\tau_i^D > 0$ be the extra cost of raising one dollar of external debt and $\tau_i^E > 0$ be the extra cost of raising one dollar of external equity. Then the firm chooses investment to maximize the enterprise value, net of the real investment and external financing costs:

$$\max_{I_i \geq 0} \frac{V_i(I_i)}{1 + r_i^A(\phi_i)} - I_i - \tau_i^D \min\{(I_i - W_i)^+, \bar{D}_i\} - \tau_i^E (I_i - W_i - \bar{D}_i)^+, \quad (12)$$

where $r_i^A(\phi_i)$ is the firm's weighted average cost of capital.

Proposition 4 (HFT can reduce equity issuance and firm investment) *Suppose the firm is in the external-equity region, so that $I_i^* > W_i + \bar{D}_i$. Assume the cost of debt and the firm's capital structure are both fixed so that*

$$\frac{\partial r_i^A(\phi_i)}{\partial \phi_i} = w_i^E \frac{\partial r_i^E(\phi_i)}{\partial \phi_i} > 0. \quad (13)$$

Then higher HFT intensity leads to lower firm investment and equity issuance:

$$\frac{\partial I_i^*}{\partial \phi_i} < 0, \quad \frac{\partial E_i^*}{\partial \phi_i} < 0.$$

The effect is concentrated among equity-reliant firms, i.e., firms with low non-equity financing capacity $W_i + \bar{D}_i$, because these firms are more likely to operate in the external-equity region.

Proof. In the external-equity region $I_i > W_i + \bar{D}_i$, the objective in (12) is differentiable and the first-order condition is $V_i'(I_i^*) = (1 + r_i^A(\phi_i))(1 + \tau_i^E)$. Since $V_i'(I_i) = \theta_i - g_i I_i$, the interior solution is

$$I_i^* = \frac{\theta_i - (1 + \tau_i^E)(1 + r_i^A(\phi_i))}{g_i}.$$

Differentiating with respect to ϕ_i gives

$$\frac{\partial I_i^*}{\partial \phi_i} = -\frac{1 + \tau_i^E}{g_i} \frac{\partial r_i^A(\phi_i)}{\partial \phi_i}.$$

Under condition (13), this derivative is negative. When $E_i^* = I_i^* - W_i - \bar{D}_i > 0$ in the external equity region, since $W_i + \bar{D}_i$ is fixed, we also have

$$\frac{\partial E_i^*}{\partial \phi_i} = \frac{\partial I_i^*}{\partial \phi_i} < 0.$$

Finally, firms with lower non-equity financing capacity $W_i + \bar{D}_i$ are more likely to satisfy $I_i^* > W_i + \bar{D}_i$, so the effect is stronger among equity-reliant firms. ■

Combining this financing block with the HFT pricing block, we can see that the corporate

Model implications	Empirical counterparts
HFT $\uparrow \Rightarrow$ required equity return / cost of capital $\uparrow$	Table 5 (ICC measures)
HFT $\uparrow \Rightarrow$ return comovement $\uparrow$	Table 6
HFT $\uparrow \Rightarrow$ investment $\downarrow$	Table 2; Table 4; Table 8
HFT $\uparrow \Rightarrow$ equity issuance $\downarrow$	Table 3; Table 4 (equity-reliant)
Debt issuance response weak/zero	Table 3
Sales growth weaker; ROA not necessarily	Table 7

financing effect is prominent when the return-comovement and liquidity-comovement costs dominate the negative average liquidity-level term such that $\frac{\partial r_i^E}{\partial \phi_i} > 0$. The key empirical implications of this model are tabulated here and correspond to our main empirical results.

3 Data and Sample

3.1 Data Sources

We obtain firm financial statement information from Compustat Global. To measure high-frequency trading (HFT) activity and obtain limit order book information, we use transaction and order book data from the Securities Industry Research Centre of Asia-Pacific (SIRCA). Our sample period spans 2000–2022.

SIRCA provides full order book event and trade data for the Australian Securities Exchange (ASX), a market that is relatively unfragmented and operates as a continuous limit order book with price–time priority, opening auctions, and closing auctions. The data contain the stock symbol, millisecond time stamp, order price and size, an order identifier, event type (submit, trade, amend, cancel), and a broker identifier. The order identifier allows us to link amendments, executions, and cancellations to the original order and to reconstruct the limit order book. SIRCA assigns broker identifiers to trader categories, including proprietary HFT firms, institutions, and retail traders. These classifications allow us to quantify HFT activity directly. Using the broker identifiers and limit order information, we compute each stock’s daily HFT trading volume share. We then aggregate the resulting measures to the

stock (firm)–year level. We provide a more detailed description of the SIRCA data in Online Appendix B.

We then merge the aggregated HFT measure with the Compustat Global sample. We exclude firms in financials and utilities and remove stocks with an annual average adjusted closing price from Compustat below \$1.00. The resulting sample used for the summary statistics contains 238 firms and 3,157 firm-year observations over 2000–2022.⁷

3.2 Summary Statistics

Table 1 reports descriptive statistics for the Australian firm-year sample used in our ASX-ITCH analyses. The sample covers 2000–2022 and contains 3,157 firm-year observations from 238 unique firms.

[Insert Table 1 about here]

The corporate investment variables exhibit substantial cross-sectional dispersion. Capital expenditures (*Capex*) average 8.728% of lagged assets, with a median of 4.410%. These statistics indicate a right-skewed distribution in which a subset of firms invests aggressively. The annual change in property, plant, and equipment scaled by lagged assets (ΔPPE) has a mean of 5.283% and a median of 1.000%. This pattern is consistent with lumpy investment and heterogeneous growth opportunities across firms.

External financing activity is also sizable and highly variable. Equity issuance (*Equity Issues*) has a mean of 10.68 and a median of 0.033. These statistics imply that many firm-years involve little or no net equity issuance, while issuance is large in the upper tail. In contrast, debt issuance (*Debt Issues*) is smaller on average, with a mean of 3.108.

The HFT measure suggests that fast trading is economically meaningful for a nontrivial subset of Australian equities. The average HFT trading-volume share is 3.683%, and the median is 3.152%. It reaches 10.27% at the 95th percentile. These statistics indicate that

⁷Although the ASX lists more than 2,000 firms, many are very small or micro-cap; the firms in our sample collectively account for more than 90% of total Australian equity market capitalization.

HFT activity concentrates in a relatively small set of stocks. Among observations with a defined top- or bottom-tercile treatment classification, the treated-group indicator has a mean of 0.546.

4 Identification Strategy: The Australian Market Setting

4.1 The Institutional Details of ASX and ASX-ITCH

The ASX introduced ASX-ITCH on April 2, 2012. It is an ultralow-latency market-data protocol that provides message-level information on ASX trading activity. All market participants can access it for a monthly fee. ITCH focuses on speed: relative to the ASX’s prior premium data products, the feed materially reduces dissemination latency. Subscription is optional, and the ASX does not disclose the identities of subscribing brokers. We therefore do not directly observe adoption. Nonetheless, traders with the strongest demand for speed should adopt first.

This institutional change is central to our study because HFT operates through rapid order submission, cancellation, and revision in the limit order book. The ITCH message stream records new orders, cancellations, modifications, and executions with millisecond timestamps. It allows us to reconstruct the order book and measure HFT activity at a granular level of detail. In addition, the ASX offers an unusually clean setting for studying HFT. Unlike highly fragmented markets such as the US and European markets, Australian equities trade in a comparatively centralized environment with transparent price–time priority. This institutional structure limits confounding from off-exchange internalization or cross-venue routing. It also makes changes in trading behavior easier and less noisy to interpret through a single order book. Finally, ASX-ITCH represents a discrete change in the market’s information and technology environment. It therefore provides a clean empirical setting to study

how fast trading evolves and influences firms' subsequent real decisions.

4.2 The Identification and Empirical Results

To implement the ASX-ITCH setting, we estimate the following difference-in-differences regression specification:

$$Y_{i,t} = \alpha + \beta \textit{Treated}_i + \theta \textit{Post}_t + \gamma (\textit{Treated}_i \times \textit{Post}_t) + \delta' X_{i,t} + \mu_i + \lambda_t + \varepsilon_{i,t}, \quad (14)$$

where $Y_{i,t}$ denotes our key variables of interest, $\textit{Cape}x_{i,t}$ and $\Delta \textit{PPE}_{i,t}$, for firm i in year t . $\textit{Post}_t$ is an indicator equal to one for observations after the 2012 introduction of ITCH. We exclude the 2012 event year from the static DiD regressions. $\textit{Treated}_i$ is an indicator for stocks with higher ex-ante exposure to high-frequency trading. The vector $X_{i,t}$ captures the full set of time-varying firm controls. The terms μ_i and λ_t denote firm and year fixed effects. Year fixed effects absorb $\textit{Post}_t$, while firm fixed effects absorb $\textit{Treated}_i$. We therefore do not identify these main effects separately. The interaction term $\textit{Treated}_i \times \textit{Post}_t$ captures the average treatment effect.

In addition, we include size-group $\times$ SIC2 fixed effects to flexibly absorb time-invariant differences across firms that operate in the same industry but fall into different size groups. This adjustment is particularly important in Australia, where firms differ substantially in size and related characteristics, including financing capacity, analyst coverage, and institutional ownership. These interacted fixed effects help ensure that persistent size- and industry-related heterogeneity does not drive our estimates. The control variables in $X_{i,t}$ include standard determinants of corporate investment and financing activity, such as firm size, profitability, cash flow, and leverage. They also include additional characteristics used throughout the paper. Online Appendix A provides the full list and detailed definitions.⁸

A key empirical task in implementing this difference-in-differences design is to define

⁸Following the literature, we lag firm size, profitability, and leverage, while we measure cash flow contemporaneously.

appropriate treated and control groups. Following the market microstructure literature, we sort firms based on their ex-ante exposure to high-frequency trading (HFT). Specifically, we compute each stock’s annual average HFT trading-volume share—HFT trading volume divided by total trading volume—during 2010 and 2011, immediately before ASX-ITCH. We use two pre-event years so that the exposure measure reflects persistent differences across firms rather than transitory fluctuations in trading activity. We then rank stocks by this pre-ITCH exposure measure and form terciles. We assign stocks in the top tercile to the treated group, while stocks in the bottom tercile form the control group.⁹ The intuition is that ASX-ITCH primarily increases the value of speed and low-latency market data. A further increase in speed should therefore matter most for stocks with the highest pre-event HFT presence. These are precisely the stocks in which speed-sensitive strategies are already most active. Figure 1 provides visual evidence consistent with this identification strategy. For high-exposure treated stocks, the HFT trading-volume share increases from less than 7% before ASX-ITCH to approximately 10% after ITCH. Low-exposure control stocks remain roughly flat at around 1%. This evidence supports the view that our treated group is the market segment most affected by ASX-ITCH.

[Insert Figure 1 about here]

Table 2 presents our difference-in-differences estimates of the effect of ASX-ITCH on corporate investment. The key coefficient is γ on $Treated \times Post$. It captures the post-2012 change in investment for treated firms relative to control firms. In Columns (1)–(2), the coefficient is -1.486 ($t = -1.900$) for Capex and -2.870 ($t = -2.450$) for ΔPPE . Thus, the baseline estimates indicate an economically meaningful contraction in both investment flows and capital-stock growth among treated firms after ASX-ITCH.

[Insert Table 2 about here]

⁹Our findings are not sensitive to the exact cutoff used to define treated and control groups.

Figure 2 provides complementary descriptive evidence on investment dynamics. Before ASX-ITCH, treated and control firms move broadly in parallel in both Capex and ΔPPE , although their average levels are not identical. After the event, the treated series falls relative to the control series, and the treated–control gap becomes more negative. The figure therefore illustrates the source of the DiD estimates in Table 2. Identification comes from the change in the between-group gap after ITCH, rather than from a comparison of unconditional investment levels.

[Insert Figure 2 about here]

Treated and control firms may differ systematically in pre-ITCH firm characteristics. To address this concern, we estimate a propensity-score model using only pre-ITCH observations. We model treatment status as a function of SIC2 industry indicators, lagged log market capitalization, and lagged log trading volume. We bound the estimated propensity scores between 0.01 and 0.99. We then construct inverse-probability weights equal to $1/\hat{p}_i$ for treated observations and $1/(1 - \hat{p}_i)$ for control observations. Next, we average these weights within firms over the pre-treatment period. We apply the resulting firm-level weight to every observation for that firm in the weighted DiD regressions. This procedure gives greater influence to treated and control firms with comparable ex-ante characteristics. It also reduces the influence of observations with treatment probabilities close to zero or one.

Figure 3 evaluates covariate balance in the year immediately before ASX-ITCH. The figure reports the weighted means of the principal firm characteristics for treated and control firms. It also reports the treated–control difference and its t -statistic. The weighted groups are statistically similar across the reported fundamentals, apart from firm size and a marginal difference in debt issuance. Because the regressions also absorb size-group $\times$ SIC2 fixed effects, they flexibly control the remaining size difference within industry-size cells.

[Insert Figure 3 about here]

Columns (3)–(4) of Table 2 report the corresponding IPW DiD estimates using these weights (see, e.g., Abadie, 2005; Acemoglu, Naidu, Restrepo, and Robinson, 2019). The weighted coefficients are -1.857 ($t = -2.120$) for Capex and -3.536 ($t = -2.610$) for Δ PPE. Both remain statistically significant and are somewhat larger in absolute magnitude than the baseline estimates. The similarity across the unweighted and reweighted specifications indicates that the firm characteristics used to predict treatment assignment do not explain the investment results.

Columns (5)–(6) use a continuous treatment measure based on ex-ante HFT exposure rather than the tercile-based treatment indicator. This specification retains the full cross-sectional variation in pre-ITCH HFT intensity. It also does not depend on the cutoff used to define treated and control firms. The coefficients on $HFT\ Exposure \times Post$ are -0.271 ($t = -2.280$) for Capex and -0.451 ($t = -2.660$) for Δ PPE. Taken together, the baseline, IPW, and continuous-exposure estimates suggest that firm characteristics and heterogeneity in firms’ exposure to HFT are unlikely to drive the investment results.¹⁰

5 Mechanisms

We now turn to the question of what drives the decline in investment. We first focus on a financing-based mechanism. Under this mechanism, the ITCH-induced increase in HFT activity raises the marginal cost of equity capital and reduces firms’ effective access to external equity.

Table 3 presents the external financing results. In the baseline specification, the DiD coefficient for Equity Issues is -6.835 ($t = -2.730$). This estimate implies that treated firms reduce net equity issuance by 6.835 percentage points more than control firms after ITCH. The corresponding coefficient for Debt Issues is statistically insignificant. Thus, the financing response concentrates in equity rather than debt. Columns (3)–(4) show that this distinction

¹⁰Online Appendix Table F1 reports estimates using restricted event windows. The coefficients for Δ PPE and Equity Issues remain negative and statistically significant in both the $[-5, +5]$ and $[-10, +10]$ windows. The Capex coefficient is statistically significant only in the wider window.

survives reweighting. The IPW estimate for Equity Issues is -5.918 ($t = -2.240$), whereas the estimate for Debt Issues remains statistically insignificant. Finally, the continuous-exposure specification produces an Equity Issues coefficient of -0.817 ($t = -2.840$). The Debt Issues coefficient is again statistically insignificant. The same equity-specific pattern therefore appears in the baseline, IPW, and continuous-treatment designs.

[Insert Table 3 about here]

Table 4 provides cross-sectional evidence consistent with this mechanism. We split firms at the median of their ex-ante three-year average equity issuance. We determine the classification before ITCH so that the shock cannot affect it. Among high-reliance firms, the post-ITCH estimates are -5.190 for Capex ($t = -2.630$), -6.852 for ΔPPE ($t = -2.590$), and -30.168 for Equity Issues ($t = -3.740$). In the low-reliance subsample, the Capex and Equity Issues coefficients are statistically insignificant. Only ΔPPE declines significantly, and its magnitude is considerably smaller at -2.482 ($t = -2.150$). Thus, firms that relied more heavily on external equity before ASX-ITCH experience the strongest reductions in investment and equity issuance.

[Insert Table 4 about here]

The evidence in Tables 3 and 4 suggests a possible external financing-cost channel. Treated firms reduce equity issuance, particularly when they enter the event with greater reliance on equity markets. To examine this mechanism more directly, we construct two implied cost-of-equity measures, r_{Gordon} and r_{OJ} . We follow Gordon and Gordon (1997) and Ohlson and Juettner-Nauroth (2005), as implemented by Gode and Mohanram (2003), respectively. We also use the earnings-forecasting approach of Hou, van Dijk, and Zhang (2012). Online Appendix C provides details. Table 5 reports positive and statistically significant DiD estimates for both measures. In the baseline specifications, the coefficients are 0.042 ($t = 5.530$) for r_{Gordon} and 0.054 ($t = 4.310$) for r_{OJ} . These estimates equal increases of 4.2 and 5.4 percentage points, or 420 and 540 basis points, in the annual required

return on equity. We then apply the same firm-level IPW weights used in Table 2. The estimates are 0.044 ($t = 5.720$) and 0.061 ($t = 4.520$), equivalent to 4.4 and 6.1 percentage points. These economically meaningful increases raise firms' equity-financing hurdle rates. A prospective project must generate a higher expected return to attract external equity capital. The estimates remain stable across the two implied-return models and across unweighted and weighted samples. This stability reduces the likelihood that one valuation formula or treatment selection based on firm characteristics drives the result.

[Insert Table 5 about here]

If the investment decline reflects tighter equity-financing capacity, treated firms may be unable to fund growth opportunities. This mechanism predicts weaker sales growth without an offsetting improvement in profitability. By contrast, managers may use more informative prices to eliminate inefficient projects. This mechanism predicts that lower investment should improve operating efficiency. We use these contrasting predictions to interpret the operating-performance results reported later in the paper.

6 Internal Validity of the ASX-ITCH Setting

6.1 Dynamic DiD and Randomized Placebo Tests

The parallel-trends assumption requires treated and control firms to have followed similar outcome paths absent ASX-ITCH. We assess its pre-event implication by testing whether the two groups exhibit differential trends before the event. Figure 4 presents this test using the IPW sample. We replace the single $Treated \times Post$ indicator with event-time interactions for years $t = -4$ through $t = 4$. We normalize $t = -1$ to zero. We otherwise retain the same outcome-specific controls, firm fixed effects, year fixed effects, size-group $\times$ SIC2 fixed effects, firm-level IPW weights, and clustered standard errors used in the main specifications.

All pre-event coefficients are statistically insignificant at the 10% level. The implied cost-of-equity estimates appear to rise gradually before ITCH, but their confidence intervals include zero. Overall, the estimates provide no evidence of differential pre-trends and support the parallel-trends assumption. After ITCH, Capex and Δ PPE move distinctly lower, and Equity Issues also fall. The implied cost of equity moves above its pre-event values and reaches its largest estimate two years after ITCH. These post-event paths are consistent with the investment, financing, and cost-of-equity results in the main tables.

[Insert Figure 4 about here]

Parallel pre-event paths address differential trends, but they do not by themselves rule out another possibility. Other persistent characteristics of the selected high-HFT firms could mechanically generate unusual estimates. We therefore conduct randomized treatment-assignment placebo tests using the same IPW analysis sample, reported in Figure 5. In each of 500 repetitions, we randomly reassign treated status across firms while preserving the actual number of treated firms. We then re-estimate the same outcome-specific DiD regression with the corresponding controls, fixed effects, and firm-level clustering used in the IPW analysis. The IPW sample remains fixed across repetitions. Only the identity of the pseudo-treated firms changes. This procedure generates the coefficient distribution that would arise if the identity of the firms most exposed to ASX-ITCH were arbitrary rather than economically meaningful.

The placebo distributions center close to zero, whereas the actual estimates lie in their tails. The one-sided randomization p -values are 0.052 for Capex, 0.018 for Δ PPE, 0.002 for Equity Issues, and 0.002 for Return Comovement. Section 7 examines the last market-quality outcome. These results imply that random treatment labels rarely reproduce the observed investment decline, contraction in equity issuance, and increase in return comovement. The dynamic DiD evidence addresses differential pre-trends, while the placebo tests show that the observed assignment of firms to high and low pre-ITCH HFT exposure is economically meaningful.

[Insert Figure 5 about here]

7 Drivers of the Increased Cost of Equity

Having established that the implied cost of equity rises after ASX-ITCH, we next examine what drives this increase. We distinguish between three possibilities. Under the liquidity channel, wider spreads raise trading and financing costs. Under the price-efficiency channel, changes in the speed of information incorporation affect the usefulness of prices. Under the common-risk channel, greater return comovement increases firms' exposure to systematic variation.

Table 6 considers three corresponding outcomes. Following Hou and Moskowitz (2005), *Delay* measures the speed with which returns incorporate information. Following Morek, Yeung, and Yu (2000) and Malceniece, Malcenieks, and Putniņš (2019), Return Comovement uses the market-model R^2 to measure the share of return variation associated with common factors. The bid-ask spread captures trading costs and stock-market liquidity. Each regression uses the market-quality controls appropriate for these outcomes: lagged log market capitalization, lagged log price, and lagged volatility. These controls differ from the corporate-outcome controls used in Tables 2–5.

The baseline coefficient for Return Comovement is 0.300 ($t = 3.180$), and the IPW estimate is similar at 0.279 ($t = 2.410$). Return Comovement equals $\ln[R^2/(1 - R^2)]$. These coefficients imply that the ratio of market-explained to firm-specific return variation rises by approximately 35% in the baseline specification and 32% in the IPW specification. Treated stocks therefore become materially more exposed to common market movements after ITCH. This pattern indicates greater exposure to systematic risk. In untabulated variance-decomposition tests, idiosyncratic return variance also declines significantly after ITCH. This finding helps explain the rise in R^2 -based comovement and indicates less firm-specific variation in treated stocks' returns. In contrast, the coefficients for Delay and Bid-ask

Spread are statistically insignificant in both specifications. Thus, a deterioration in average spread liquidity is unlikely to explain the higher cost of equity. We also find no evidence that ITCH changes the speed with which prices incorporate information. Instead, the larger common component in treated-stock returns is consistent with a risk-based explanation for the higher required returns reported in Table 5.

[Insert Table 6 about here]

8 Additional Analysis and Robustness

8.1 Operating Performance and the Financing-Cost Channel

Operating performance provides a complementary way to distinguish the financing-cost channel from improved managerial learning. Tighter equity-financing capacity may prevent treated firms from funding otherwise valuable growth opportunities. Under this mechanism, the investment contraction may lead to weaker sales growth without an improvement in profitability. In contrast, managers may learn from more informative prices and deliberately eliminate wasteful projects. Under this mechanism, lower investment should improve operating efficiency and profitability.

Table 7 reports DiD estimates for return on assets (ROA) and sales growth. The ROA coefficient is statistically insignificant in the baseline, IPW, and continuous-exposure specifications. Thus, the post-ITCH investment decline does not coincide with higher accounting profitability. Sales growth falls by 13.079 percentage points in the baseline specification ($t = -2.530$) and by 11.490 percentage points in the IPW specification ($t = -1.950$). The continuous-exposure estimate is also negative at -1.252 ($t = -1.830$). This combination of unchanged profitability and weaker sales growth is consistent with firms scaling back expansion because equity financing becomes more costly. It also provides indirect evidence against the managerial-learning interpretation. The pattern is more consistent with tighter financing capacity than with an improvement in managerial learning.

[Insert Table 7 about here]

8.2 Alternative Setting: Exchange Co-location

Our main identification strategy exploits the introduction of ASX-ITCH within a single market. As an alternative setting, we examine the staggered rollout of exchange co-location services across markets. Co-location allows traders to place their servers in close physical proximity to an exchange’s matching engine. This proximity reduces latency and increases the value of speed. Exchanges introduced co-location at different times. The staggered rollout therefore provides additional variation distinct from the ASX-ITCH institutional setting.

The co-location setting has important limitations. Exchanges choose when to introduce co-location, so adoption is not a purely exogenous shock. It may respond to trading demand, competitive pressure, or broader technology investments. It may also coincide with other market reforms. Cross-country institutional differences make this variation less tightly controlled than the single-market ASX-ITCH setting. We therefore use co-location as corroborating evidence from a distinct latency-reducing market change. We do not treat it as a substitute for our main ASX-ITCH design.

Aitken, Cumming, and Zhan (2023) document substantial differences among co-location dates reported in prior studies, including those in Boehmer, Fong, and Wu (2021), and show that event-date errors can materially affect empirical inference. We therefore begin with 15 candidate events from Boehmer, Fong, and Wu (2021) and Aitken, Cumming, and Zhan (2023). We compare the dates across the two studies and verify each candidate using exchange press releases and contemporaneous news reports. When the dates reported in the two studies conflict, we retain the event only if these independent sources resolve the discrepancy. We exclude events with unavailable dates or unresolved ambiguity. This procedure produces a final list of ten events. The list is smaller than that in Boehmer, Fong, and Wu (2021), but it reduces event-time misclassification and improves the accuracy of the

co-location design. Online Appendix E reports the included exchanges and dates.

We then estimate a staggered DiD specification. The key regressor, *Co-location*, equals one for firm-year observations after the stock’s listing exchange begins to offer co-location services. We exclude observations in the event year to allow trading technology and market participation to adjust.

Table 8 reports two complementary estimates. Panel A presents the staggered DiD specification. It uses the ten included co-location events, excludes the adoption year, limits each cohort to an event window of $[-10, +10]$, and includes firm and year fixed effects, Country $\times$ SIC2 fixed effects, and the controls reported in the table. Co-location is associated with statistically significant declines in Capex (-0.508 , $t = -4.407$), Δ PPE (-0.823 , $t = -5.553$), and Equity Issues (-0.798 , $t = -3.721$). It is also associated with higher implied costs of equity under both the Gordon and Ohlson–Juettner measures. The Debt Issues estimate is statistically insignificant.

Panel B reports a stacked DiD design. A single pooled staggered coefficient can mix comparisons across adoption cohorts and event horizons, which makes the estimate sensitive to treatment-effect heterogeneity (de Chaisemartin and D’Haultfoeuille, 2020; Borusyak, Jaravel, and Spiess, 2024). Related concerns also arise in continuous-treatment DiD settings (Callaway, Goodman-Bacon, and Sant’Anna, 2024). Following the event-specific stacking logic in Cengiz, Dube, Lindner, and Zipperer (2019), we separate the adoption cohorts. The stacked design compares each cohort only with later cohorts while those firms remain untreated and removes a control firm when its own exchange adopts co-location. Stack $\times$ Firm fixed effects allow the same control firm to enter different cohort-specific comparisons without imposing a common firm baseline across stacks. Stack $\times$ Year fixed effects absorb shocks within each comparison. We also estimate propensity scores from pre-adoption characteristics and apply the resulting weights within each stack. This design provides a more robust estimate of the average co-location effect than a single pooled staggered coefficient.

The Panel B results support the central findings. Co-location reduces Capex by 0.830

percentage points ($t = -1.861$), Δ PPE by 2.146 percentage points ($t = -4.229$), and Equity Issues by 1.522 percentage points ($t = -2.152$). The Ohlson–Juettner cost of equity increases by 0.032 ($t = 3.581$). The Debt Issues and Gordon cost-of-equity estimates are statistically insignificant. Thus, cohort composition, already-treated comparisons, and firm characteristics do not explain the investment and equity-issuance findings. Figure F3 reports the corresponding dynamic estimates. Several pre-adoption coefficients differ statistically from zero, so we do not interpret these dynamics as evidence of parallel trends. This limitation reinforces our treatment of co-location as corroborating evidence rather than a standalone natural experiment.

[Insert Table 8 about here]

8.3 Alternative Explanation: Managerial Learning and Allocation Efficiency

One alternative interpretation of our finding is a managerial-learning, or allocation-efficiency, channel. In this view, the ITCH-induced increase in HFT improves the information environment by accelerating the incorporation of firm- and market-level news into prices. A post-ITCH decline in average investment could therefore reflect improved investment discipline rather than tighter financing conditions. Firms may learn that some marginal projects have lower expected profitability than previously believed and optimally reduce capital expenditures.

This explanation has a clear empirical implication. If ITCH improves the information content of prices in a way that managers use to guide real decisions, firms’ investment should become more responsive to measures of investment opportunities. We therefore augment the baseline specifications with lagged Tobin’s q and its interactions with *Post* and treatment status. This specification allows the investment– q slope to change differentially for treated firms after ITCH. Under the managerial-learning hypothesis, the triple-interaction coefficient

should be positive and statistically significant.

Online Appendix Table F2 reports the results. In the IPW ASX-ITCH specifications, the post-ITCH change in investment- q sensitivity is statistically insignificant for both outcomes. In the co-location setting, the Q interaction is negative and statistically significant for Capex and statistically insignificant for Δ PPE. Neither setting shows a positive differential increase in investment- q sensitivity. Table 6 also reports statistically insignificant Delay estimates. Together, this evidence provides no support for improved managerial learning as the explanation for the investment decline. The results are more consistent with the financing-capacity mechanism. Under this mechanism, a higher required return on equity reduces firms' ability to finance marginal investment projects.

9 Conclusion

This paper studies whether financial market innovation such as high-frequency trading affects firms' financing and investment decisions. We exploit the introduction of ASX-ITCH in April 2012, and use order-level data and broker identifiers to directly measure HFT activity and HFT exposure. Our difference-in-differences estimates show that firms with greater HFT exposure reduce investment and equity issuance after ITCH. The effects are strongest among firms that relied more heavily on external equity. The magnitudes are economically meaningful and persist across the various specifications.

The evidence points to a financing-cost channel. The implied cost of equity rises after ITCH, while equity issuance falls. Return comovement also rises, consistent with greater exposure to common risk. We find no evidence that ITCH changes price delay or bid-ask spreads. We also find no evidence of a change in investment- q sensitivity. The operating-performance results provide no indication that firms become more profitable after reducing investment. The results therefore do not support liquidity, price-efficiency, or managerial-learning explanations for the decline in firm investment.

These results survive dynamic DiD tests and randomized placebo tests. The co-location analysis provides a distinct external check. ASX-ITCH is a market-data and trading-protocol change within one market. Co-location is an exchange-infrastructure shock that reduces physical distance to the matching engine and rolls out at different times across markets. The two shocks therefore differ in their institutional setting and need not affect HFT through identical margins. Yet both generate declines in investment and equity issuance, together with higher implied costs of equity. This convergence suggests that the corporate effects reflect broader consequences of faster trading rather than a feature unique to the ASX-ITCH event. Financial market trading speed can affect real outcomes through the cost and availability of external equity.

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Table 1
Descriptive statistics

This table reports descriptive statistics for the Australian sample used in the ASX-ITCH analysis. The sample consists of firm-year observations for 2000–2022. All variables are defined in Online Appendix A.

Variable	N	Mean	Median	SD	P5	P95
Capex	3,157	8.728	4.410	11.30	0.457	39.66
Δ PPE	2,895	5.283	1.000	12.75	-7.712	39.15
Equity Issues	3,157	10.68	0.033	34.02	0.000	66.58
Debt Issues	3,157	3.108	0.000	12.14	-12.13	30.03
Treated	2,169	0.546	1.000	0.498	0.000	1.000
HFT Exposure (%)	3,157	3.683	3.152	3.117	0.199	10.27
Market Cap	2,976	3,771	953.4	5,794	62.59	19,056
Cash Flow	3,081	10.23	11.12	18.93	-17.52	38.97
Size	3,157	5.794	5.821	1.885	2.661	8.574
Past Profitability	3,152	13.49	14.33	21.26	-18.46	44.70
Leverage	3,156	23.65	22.97	19.63	0.000	59.08

Table 2
ASX-ITCH and Corporate Investment

This table reports difference-in-differences (DiD) estimates of the effect of high-frequency trading (HFT) on corporate investment around the introduction of ASX-ITCH. The dependent variables are capital expenditure (Capex) and the annual change in property, plant, and equipment scaled by lagged assets (Δ PPE), as indicated in the column headings. Columns (1)–(2) report baseline DiD estimates. Columns (3)–(4) report inverse probability weighted (IPW) DiD estimates. Propensity scores are estimated in the pre-ITCH period using SIC2 industry indicators, lagged log market capitalization, and lagged log trading volume; the resulting firm-level inverse-probability weights are applied to all firm-year observations. Columns (5)–(6) use a continuous ex-ante HFT exposure treatment measure, defined as the average HFT volume share prior to the introduction of ASX-ITCH. All variables are defined in Online Appendix A. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Baseline		IPW		HFT Intensity	
	(1) Capex	(2) Δ PPE	(3) Capex	(4) Δ PPE	(5) Capex	(6) Δ PPE
Treated $\times$ Post	-1.486* (-1.900)	-2.870** (-2.450)	-1.857** (-2.120)	-3.536*** (-2.610)		
HFT Exposure $\times$ Post					-0.271** (-2.280)	-0.451*** (-2.660)
Size	-3.062*** (-4.926)	-3.359*** (-3.871)	-2.637*** (-3.828)	-1.612* (-2.017)	-3.160*** (-6.056)	-3.634*** (-4.795)
Cash flow	0.123** (2.442)	0.170** (2.233)	0.077 (1.568)	0.150** (2.626)	0.050 (0.981)	0.039 (0.485)
Past profitability	-0.084 (-1.528)	-0.079 (-1.017)	0.053 (0.979)	0.039 (0.504)	-0.055 (-1.142)	0.003 (0.039)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	1,797	1,809	1,797	1,809	2,552	2,581
Adj. R ²	0.569	0.268	0.542	0.237	0.627	0.318

Table 3
ASX-ITCH and External Financing

This table reports difference-in-differences (DiD) estimates of the effect of high-frequency trading (HFT) on firms' external financing around the introduction of ASX-ITCH. The dependent variables are equity issues (Equity Issues) and debt issues (Debt Issues), as indicated in the column headings. Columns (1)–(2) report baseline DiD estimates, Columns (3)–(4) report inverse probability weighting (IPW) DiD estimates, and Columns (5)–(6) use a continuous ex-ante HFT exposure treatment measure, defined as the average HFT volume share prior to the introduction of ASX-ITCH. All variables are defined in Online Appendix A. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. *t*-statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Baseline		IPW		HFT Intensity	
	(1) Equity Issues	(2) Debt Issues	(3) Equity Issues	(4) Debt Issues	(5) Equity Issues	(6) Debt Issues
Treated $\times$ Post	-6.835*** (-2.730)	-1.934 (-1.400)	-5.918** (-2.240)	-1.793 (-1.110)		
HFT Exposure $\times$ Post					-0.817*** (-2.840)	-0.236 (-1.270)
Size	-7.764*** (-4.558)	-1.564** (-2.413)	-5.872*** (-3.162)	-1.643 (-1.494)	-8.905*** (-5.673)	-1.297** (-2.503)
Cash Flow	0.147 (0.835)	-0.080 (-0.641)	0.216 (1.157)	-0.077 (-1.265)	-0.079 (-0.442)	-0.071 (-0.846)
Past Profitability	-0.451** (-2.311)	0.122 (0.989)	-0.011 (-0.057)	0.323** (2.578)	-0.263 (-1.462)	0.090 (1.056)
Leverage	0.039 (0.866)	-0.205*** (-8.729)	0.092 (1.456)	-0.238*** (-8.332)	0.035 (0.943)	-0.222*** (-8.513)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	1,800	1,809	1,800	1,809	2,559	2,581
Adj. R ²	0.315	0.122	0.191	0.206	0.429	0.123

Table 4
Cross-sectional Results by Equity Financing Reliance

This table reports difference-in-differences (DiD) estimates examining cross-sectional heterogeneity in the effect of high-frequency trading (HFT) around the introduction of ASX-ITCH. The dependent variables are capital expenditure (Capex), the annual change in property, plant, and equipment scaled by lagged assets (Δ PPE), and equity issues (Equity Issues), as indicated in the column headings. Columns (1)–(3) and Columns (4)–(6) report results for subsamples with high versus low equity financing reliance. All variables are defined in Online Appendix A. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	High Equity Reliance			Low Equity Reliance		
	(1) Capex	(2) Δ PPE	(3) Equity Issues	(4) Capex	(5) Δ PPE	(6) Equity Issues
Treated $\times$ Post	-5.190** (-2.630)	-6.852** (-2.590)	-30.168*** (-3.740)	-0.455 (-0.500)	-2.482** (-2.150)	-0.012 (-0.010)
Size	-3.462*** (-4.414)	-4.138*** (-3.749)	-8.221*** (-3.087)	-2.385** (-2.825)	-2.456* (-1.879)	-5.131** (-2.384)
Cash Flow	0.214*** (7.366)	0.230** (2.503)	0.206 (1.028)	-0.053 (-0.498)	0.057 (0.864)	0.036 (0.246)
Past Profitability	-0.193*** (-8.792)	-0.160** (-2.464)	-0.551*** (-3.154)	0.104 (1.184)	0.075 (1.106)	-0.081 (-0.552)
Leverage			0.087 (0.846)			0.002 (0.072)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	546	556	548	914	916	915
Adj. R ²	0.520	0.272	0.399	0.634	0.189	0.129

Table 5
ASX-ITCH and Implied Cost of Equity

This table reports difference-in-differences (DiD) estimates of the effect of high-frequency trading (HFT) around the introduction of ASX-ITCH on firms' implied cost of equity. The dependent variables are r_{Gordon} and r_{OJ} , as indicated in the column headings. All variables are defined in Online Appendix A. Columns (1)–(2) report baseline DiD estimates, and Columns (3)–(4) report inverse probability weighting (IPW) DiD estimates. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Baseline		IPW	
	(1)	(2)	(3)	(4)
	r_{Gordon}	r_{OJ}	r_{Gordon}	r_{OJ}
Treated $\times$ Post	0.042*** (5.530)	0.054*** (4.310)	0.044*** (5.720)	0.061*** (4.520)
Size	0.011** (2.855)	0.013* (1.840)	0.010 (1.151)	0.039*** (3.255)
Cash Flow	0.002*** (3.398)	0.002* (2.057)	0.002** (2.202)	0.003*** (3.256)
Past Profitability	-0.000 (-0.886)	-0.001 (-0.986)	-0.000 (-0.866)	-0.001 (-1.468)
Leverage	0.000* (2.021)	0.000* (1.805)	0.000* (1.812)	0.000 (1.209)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes
N	1,174	652	1,174	652
Adj. R ²	0.419	0.351	0.495	0.445

Table 6
Price Informativeness, Return Comovement, and Liquidity

This table reports difference-in-differences (DiD) estimates examining the effect of high-frequency trading (HFT) around the introduction of ASX-ITCH on price informativeness, return comovement, and liquidity. The dependent variables are *Delay* (Hou and Moskowitz, 2005), *Return Comovement* (Morck, Yeung, and Yu, 2000; Malceniece, Malcenieks, and Putniņš, 2019), and *Bid-ask Spread*, as indicated in the column headings. Columns (1)–(3) report baseline DiD estimates and Columns (4)–(6) report inverse probability weighting (IPW) DiD estimates. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. *t*-statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Baseline			IPW		
	(1) Delay	(2) Comovement	(3) Bid-ask Spread	(4) Delay	(5) Comovement	(6) Bid-ask Spread
Treated $\times$ Post	-0.040 (-1.180)	0.300*** (3.180)	0.166 (0.780)	-0.040 (-1.140)	0.279** (2.410)	0.483 (1.130)
Ln(Market Cap)	-0.023 (-1.218)	0.043 (0.493)	-0.551*** (-4.640)	-0.039 (-1.442)	0.082 (0.721)	-1.152*** (-4.491)
Ln(Price)	-0.015 (-0.809)	0.198** (2.693)	-0.275* (-2.044)	-0.041 (-1.385)	0.275** (2.533)	0.013 (0.047)
Volatility	-0.396 (-0.345)	8.384** (2.132)	-43.176*** (-3.479)	-1.459 (-0.939)	15.066* (2.060)	-30.105 (-1.554)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	1,741	1,731	1,040	1,741	1,731	1,040
Adj. R ²	0.215	0.369	0.860	0.159	0.289	0.857

Table 7
ASX-ITCH and Firm Operating Performance

This table reports difference-in-differences (DiD) estimates of the effect of high-frequency trading (HFT) around the introduction of ASX-ITCH on firm operating performance. The dependent variables are return on assets (ROA) and sales growth, as indicated in the column headings. Columns (1)–(2) report baseline DiD estimates, Columns (3)–(4) report inverse probability weighting (IPW) DiD estimates, and Columns (5)–(6) use a continuous ex-ante HFT exposure treatment measure. All variables are defined in Online Appendix A. Regressions include firm and year fixed effects, as well as Size Group $\times$ SIC2 fixed effects. *t*-statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	Baseline		IPW		HFT Intensity	
	(1) ROA	(2) Sales Growth	(3) ROA	(4) Sales Growth	(5) ROA	(6) Sales Growth
Treated $\times$ Post	0.158 (0.370)	-13.079** (-2.530)	-0.025 (-0.070)	-11.490* (-1.950)		
HFT Exposure $\times$ Post					0.027 (0.510)	-1.252* (-1.830)
Size	-0.486*** (-3.129)	-19.034*** (-4.152)	-0.418* (-1.783)	-13.595*** (-3.382)	-0.385** (-2.579)	-19.332*** (-4.731)
Cash Flow	0.905*** (35.893)	0.440 (0.847)	0.889*** (29.081)	-0.444 (-1.582)	0.912*** (56.501)	0.252 (0.599)
Past Profitability	-0.046** (-2.116)	0.631 (1.079)	-0.034 (-1.206)	2.181*** (4.668)	-0.034** (-2.365)	0.725 (1.721)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	1,785	1,787	1,785	1,787	2,547	2,531
Adj. R ²	0.960	0.262	0.943	0.298	0.971	0.297

Table 8
Staggered and Stacked DiD Results from the Co-location Setting

This table presents estimates of the effect of exchange co-location on corporate investment, financing, and implied costs of equity. Panel A reports the staggered DiD specification. It uses the ten included co-location events, excludes the adoption year, and limits each event to the $[-10, +10]$ window. These regressions include firm and year fixed effects and Country $\times$ SIC2 fixed effects. Panel B reports stacked DiD estimates. For each adoption cohort, the stack includes firms treated in that cohort and firms from later cohorts only while they remain untreated. Each stack uses the $[-4, +3]$ window and excludes the adoption year. Panel B estimates propensity scores within each stack in year -1 using SIC2, lagged size, and past profitability, and applies average-treatment-effect weights trimmed at the 1st and 99th percentiles. These regressions include Stack $\times$ Firm and Stack $\times$ Year fixed effects. All specifications include the controls reported in the table. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	(1) Capex	(2) Δ PPE	(3) Equity Issues	(4) Debt Issues	(5) r_{Gordon}	(6) r_{OJ}
<i>Panel A: Staggered DiD Results</i>						
Co-location	-0.508*** (-4.407)	-0.823*** (-5.553)	-0.798*** (-3.721)	-0.319 (-1.455)	0.006** (2.546)	0.010*** (3.789)
Size	-1.203*** (-33.873)	-1.327*** (-35.871)	-2.102*** (-36.648)	-1.492*** (-25.086)	0.003*** (2.939)	-0.010*** (-8.585)
Cash Flow	0.032*** (8.978)	0.056*** (12.457)	-0.009 (-1.260)	-0.160*** (-21.577)	0.002*** (14.012)	-0.001*** (-3.494)
Past Profitability	0.094*** (23.065)	0.105*** (22.088)	0.027*** (3.539)	0.184*** (23.744)	-0.001*** (-5.996)	-0.001*** (-6.009)
Leverage			0.034*** (17.793)	-0.120*** (-54.197)	0.000*** (6.216)	0.000** (2.433)
GDP Growth	0.072*** (5.523)	0.109*** (6.755)	0.092*** (5.051)	0.010 (0.433)	-0.001*** (-3.124)	0.000 (0.648)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Country $\times$ SIC2 FE	Yes	Yes	Yes	Yes	Yes	Yes
N	132,985	141,566	134,046	141,317	47,676	39,552
Adj. R ²	0.453	0.219	0.312	0.147	0.424	0.553
<i>Panel B: Stacked DiD Results</i>						
Co-location	-0.830* (-1.861)	-2.146*** (-4.229)	-1.522** (-2.152)	-1.586 (-1.629)	0.007 (0.863)	0.032*** (3.581)
Size	-1.088*** (-9.903)	-1.637*** (-12.499)	-1.840*** (-8.538)	-2.625*** (-10.749)	0.012*** (3.607)	0.003 (0.697)
Cash Flow	0.044*** (3.888)	0.066*** (4.624)	-0.021 (-0.953)	-0.043* (-1.811)	0.003*** (7.076)	0.000 (0.159)
Past Profitability	0.085*** (6.687)	0.106*** (6.818)	0.098*** (4.386)	0.217*** (8.943)	-0.001** (-2.550)	-0.000 (-1.004)
Leverage			-0.084*** (-11.916)	0.281*** (29.067)	0.000 (0.098)	0.000 (1.303)
GDP Growth	-0.073* (-1.776)	-0.167*** (-2.928)	0.114* (1.682)	-0.127 (-1.435)	0.000 (0.021)	0.003** (2.082)
Stack $\times$ Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Stack $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes
N	106,320	112,548	107,039	111,452	39,053	31,771
Adj. R ²	0.572	0.364	0.324	0.360	0.529	0.620

Figure 1. HFT Activity around ASX-ITCH Introduction

This figure plots the monthly equal-weighted HFT trade volume share for treated and control groups over the period January 2012 to December 2012. The treated group consists of stocks in the top tercile of HFT exposure, and the control group consists of stocks in the bottom tercile, based on the average HFT trading volume scaled by total trading volume during 2010–2011. The dashed vertical line indicates April 2012, which is the month ASX introduced the ITCH protocol. The shaded area represents the gap between the two groups.

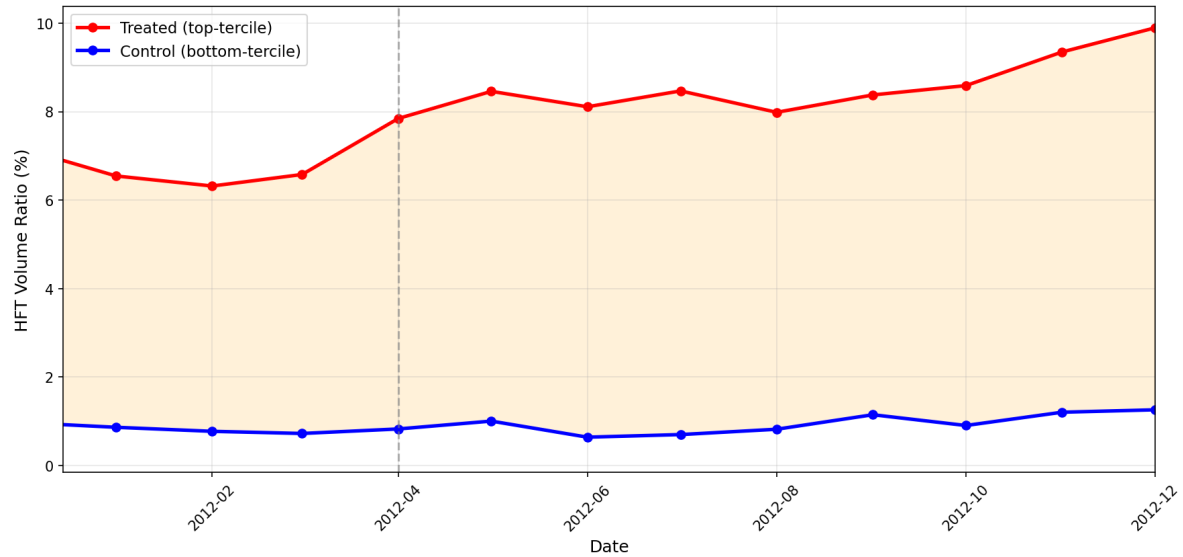


Figure 2. Corporate Investment around ASX-ITCH Introduction

This figure plots yearly mean Capex and Δ PPE for treated and control firms around the introduction of ASX-ITCH. The treated group consists of stocks in the top tercile of pre-ITCH HFT exposure, and the control group consists of stocks in the bottom tercile. For each variable, we also report the average treated-control difference in the pre period and in the post period to summarize how the gap changes after ASX-ITCH.

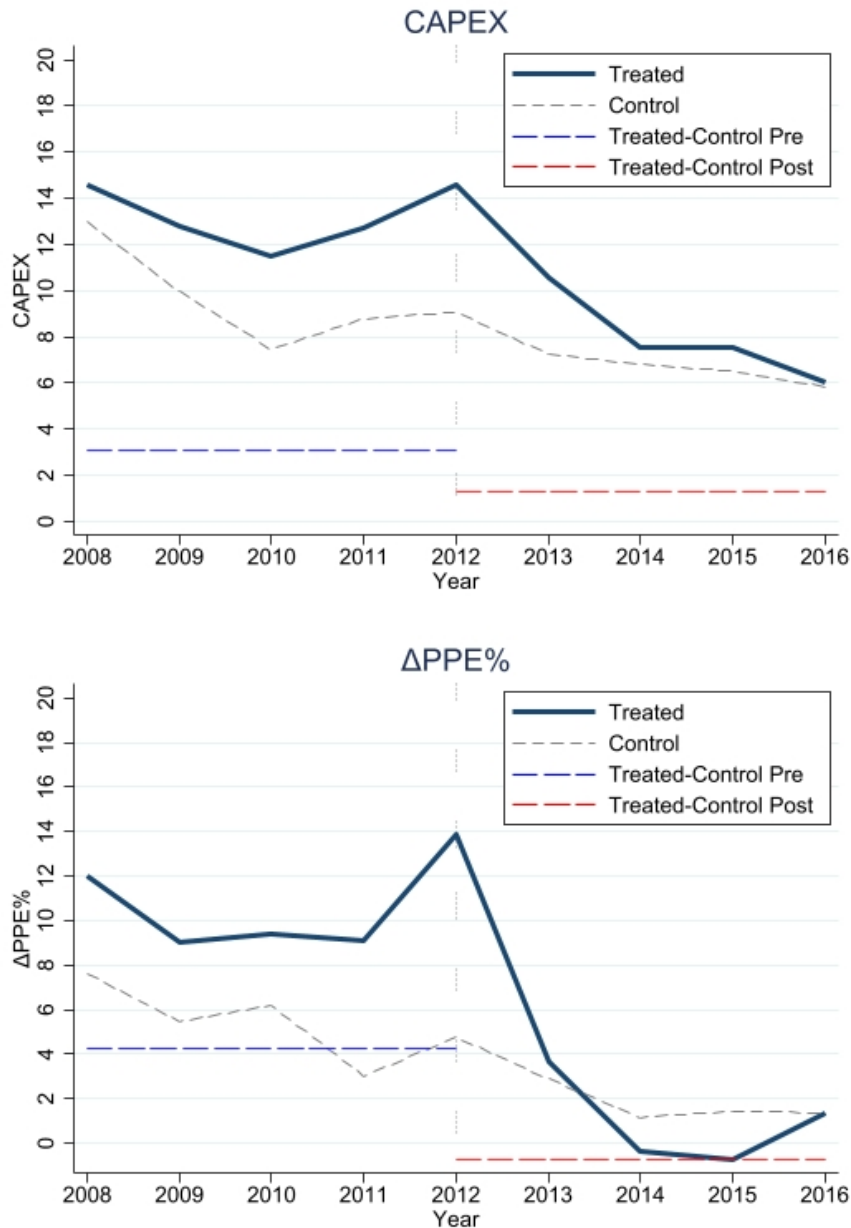


Figure 3. Covariate Balance in the IPW-Weighted Sample

This figure plots IPW-weighted means for treated and control firms in the year immediately preceding ASX-ITCH. The treated group consists of firms in the top tercile of pre-ITCH HFT exposure, and the control group consists of firms in the bottom tercile. Bars report weighted means for treated and control firms; the right-hand side reports treated-control differences and corresponding t -statistics. All variables are expressed in percentage-point-equivalent units, with Tobin Q multiplied by 100 for comparability; Size is measured as log assets.

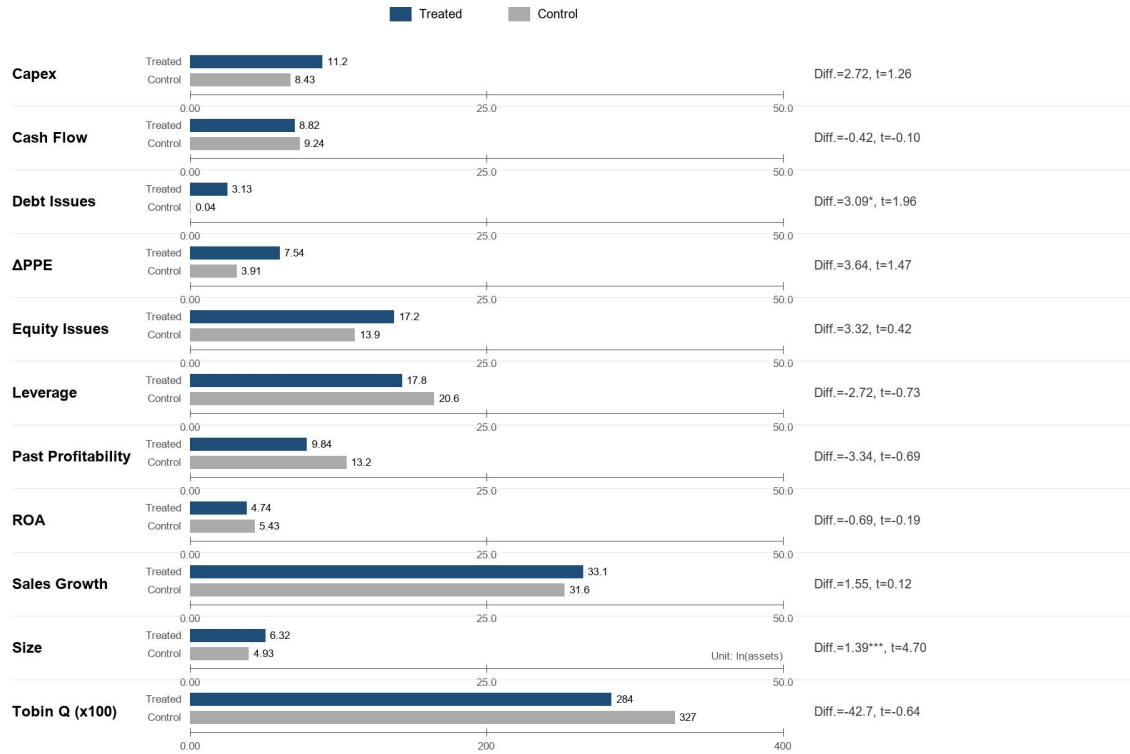


Figure 4. IPW Dynamic Difference-in-Differences Estimates around ASX-ITCH

This figure plots inverse-probability-weighted dynamic difference-in-differences estimates around the introduction of ASX-ITCH for Capex, Δ PPE, Equity Issues, and the implied cost of equity measured using r_{OJ} . Event time $t = -1$ is the omitted reference period, and the estimation window extends from $t = -4$ through $t = 4$. The regressions use the same controls and fixed effects as the corresponding IPW specifications in the main tables. Standard errors are clustered at the firm level. The light dashed vertical intervals report 90% confidence intervals, and the vertical dashed line marks the introduction of ASX-ITCH at $t = 0$.

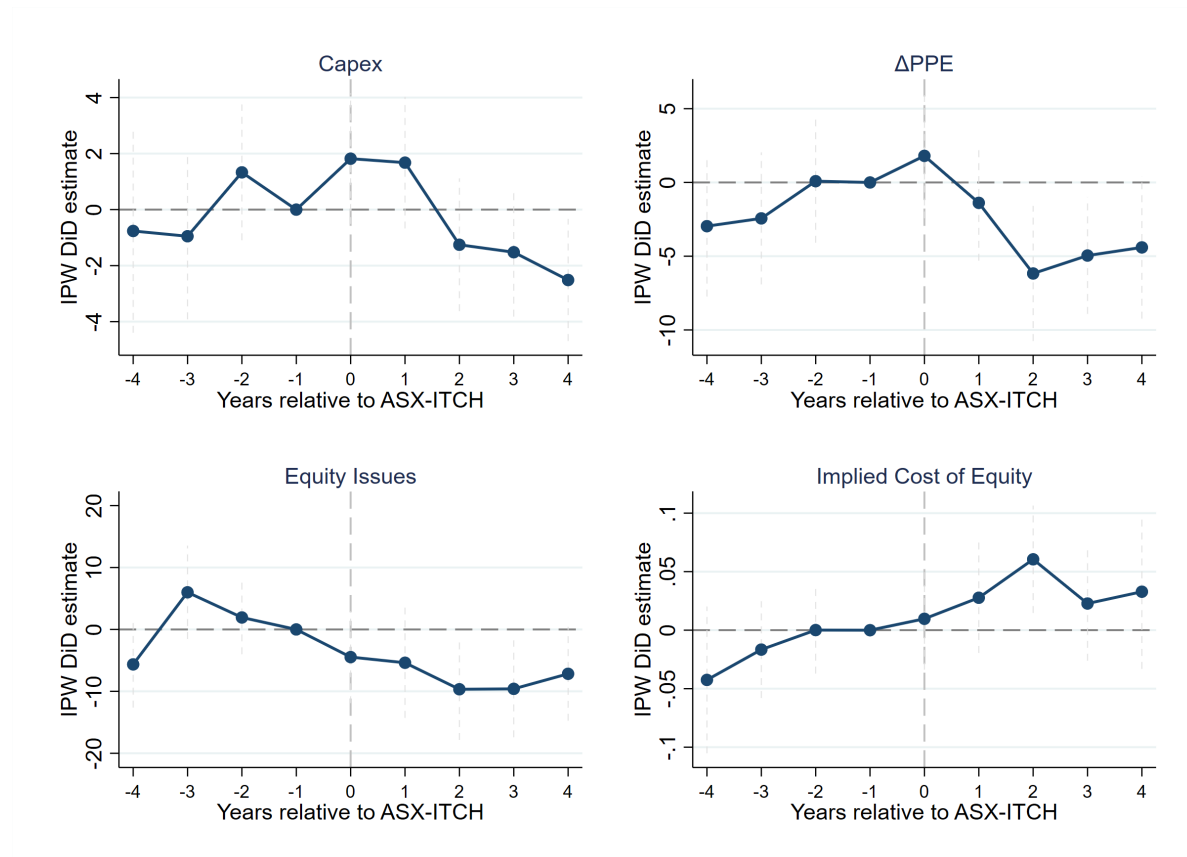
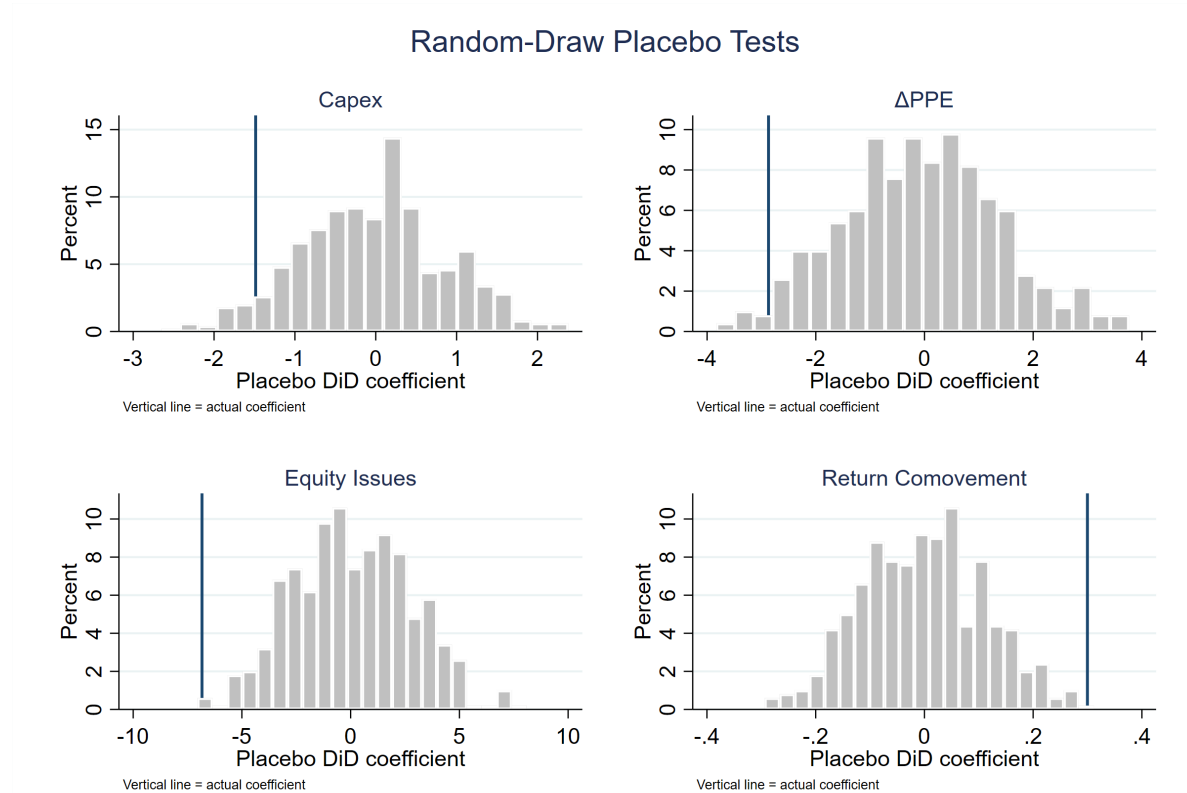


Figure 5. Randomized Treatment-Assignment Placebo Tests

This figure reports randomized treatment-assignment placebo tests for Capex, Δ PPE, Equity Issues, and Return Comovement using the same IPW analysis sample as the main tests. In each of 500 repetitions, treated status is randomly reassigned across firms while preserving the actual number of treated firms, and the corresponding outcome-specific DiD specification is re-estimated. The IPW sample remains fixed, and only the identity of the pseudo-treated firms changes. Each histogram shows the distribution of placebo DiD coefficients; the solid vertical line marks the coefficient obtained using the actual treatment assignment. The regressions include the same outcome-specific controls, firm fixed effects, year fixed effects, and Size Group $\times$ SIC2 fixed effects as the corresponding IPW specifications. Standard errors are clustered at the firm level.



Placebo treated firms are randomly drawn at the firm level while preserving the actual number of treated firms. Each histogram uses 500 repetitions.

Online Appendix for

When “Milliseconds” Matter: The Real Effects of High-Frequency Trading

A. Variable Definitions

Variables	Definitions
<i>Capex</i>	Capital expenditure divided by lagged assets, multiplied by 100.
ΔPPE	Annual change in property, plant, and equipment (PPE) divided by lagged total assets, multiplied by 100.
<i>HFT Exposure (%)</i>	Ex-ante HFT trading-volume share averaged over 2010–2011 (percent).
<i>Treated</i>	Indicator equal to one for firms in the ASX-ITCH treated group.
<i>Post</i>	Indicator equal to one for years after ASX-ITCH introduction.
<i>Co-location</i>	Indicator for years after a stock exchange allows co-location.
<i>Cash Flow</i>	Net income before extraordinary items plus depreciation, scaled by lagged assets and multiplied by 100.
<i>Equity Issues</i>	Sale of common and preferred stock divided by lagged total assets, multiplied by 100.
<i>Debt Issues</i>	Change in total debt scaled by lagged total assets, multiplied by 100.
<i>Market Cap</i>	Market capitalization in millions.
<i>Price</i>	Annual average adjusted daily closing price from Compustat Global.
<i>Volatility</i>	Annual average daily high-low volatility.
<i>Delay</i>	Price delay measure following Hou and Moskowitz (2005) .
<i>Bid-ask Spread</i>	Annual average of the daily relative bid-ask spread, multiplied by 100: $100 \times D_{it}^{-1} \sum_{d \in t} Relative\ Spread_{i,d}$, where $Relative\ Spread_{i,d} = (Ask_{i,d} - Bid_{i,d}) / [(Ask_{i,d} + Bid_{i,d}) / 2]$.
<i>Size</i>	Natural log of total assets in millions.
<i>Return Comovement</i>	Logit transformation of R^2 from the market model following (Morck, Yeung, and Yu, 2000 ; Malceniece, Malcenieks, and Putniņš, 2019).
<i>ROA</i>	Return on assets.
<i>Sales Growth</i>	Annual growth of sales (percent).
<i>Leverage</i>	Total debt divided by the sum of total debt and book equity, multiplied by 100.
<i>GDP Growth</i>	Annual real GDP growth in the firm’s headquarters country (percent).
<i>Past Profitability</i>	Operating income before depreciation divided by lagged total assets, multiplied by 100.
<i>Q</i>	Tobin’s q: market value of assets divided by book value of assets.
r_{Gordon}	Implied cost of equity from the Gordon and Gordon (1997) model.
r_{OJ}	Implied cost of equity from the Ohlson and Juettner-Nauroth (2005) model.

B. SIRCA Order-Level Data

Our intraday data come from the Securities Industry Research Centre of Asia-Pacific (SIRCA). The dataset provides complete order-level records for all securities traded on the Australian Securities Exchange (ASX), including order entries, amendments, and cancellations. Each record contains the instrument identifier, date, timestamp, order type, price, volume, and unique order identifiers (Bid ID or Ask ID). Crucially, every order record includes a *Buyer Broker ID* and a *Seller Broker ID*, which are numeric codes assigned by the ASX to each registered broker. These broker identifiers allow us to trace each order and trade to a specific broker and, using the broker-level classifications, we can categorize trading activity by trader type (e.g., high-frequency trader, institutional, or retail). Figure OA1 displays a sample of the raw data.

Figure OA1. Sample of SIRCA Order-Level Data

Notes: This figure displays a subset of the raw SIRCA order-level data for illustration. Each row represents an individual order entry on the ASX. The last two columns—Buyer Broker ID and Seller Broker ID—are numeric codes that uniquely identify the broker behind each side of the order, enabling classification of trading activity by trader type.

#	Instrument	Date	Time	Record Type	Price	Volume	Undisclosed Volume	Value	Qualifiers	Trans ID	Bid ID	Ask ID	Bid/Ask	Entry Time	Old Price	Old Volume	Buyer Broker ID	Seller Broker ID
1	AAC	20090701	00:00:00.000	ENTER	1.250	2,000	0	2,500	0	8,731,207,287,894,064,986			B				140	
2	AAC	20090701	00:00:00.000	ENTER	1.310	2,000	0	2,620	0	13,794,766,320,232,688,926			B				140	
3	AAC	20090701	00:00:00.000	ENTER	1.890	5,600	0	10,584	0		9,661,681,926,936,620,350		A					638
4	AAC	20090701	00:00:00.000	ENTER	1.240	12,000	0	14,880	0	4,380,369,974,567,319,199			B				238	
5	AAC	20090701	00:00:00.000	ENTER	1.395	2,970	0	4,143.15	0		2,469,874,737,067,806,307		A					140
6	AAC	20090701	00:00:00.000	ENTER	1.500	5,000	0	7,500	0		2,330,697,264,823,065,825		A					140
7	AAC	20090701	00:00:00.000	ENTER	2.000	5,000	0	10,000	0		16,265,744,095,415,229,378		A					406
8	AAC	20090701	00:00:00.000	ENTER	1.250	8,000	0	10,000	0	12,547,711,480,154,805,976			B				140	
9	AAC	20090701	00:00:00.000	ENTER	1.195	2,000	0	2,390	0	13,295,032,936,119,190,080			B				105	
10	AAC	20090701	00:00:00.000	ENTER	1.525	2,180	0	3,324.5	0		9,639,953,771,750,948,903		A					140
11	AAC	20090701	00:00:00.000	ENTER	1.870	1,850	0	3,459.5	0		11,707,757,566,189,275,770		A					144
12	AAC	20090701	00:00:00.000	ENTER	1.300	4,000	0	5,200	0	14,646,001,332,497,746,733			B				105	
13	AAC	20090701	00:00:00.000	ENTER	1.320	5,000	0	6,600	0	7,278,416,486,959,282,254			B					140
14	AAC	20090701	00:00:00.000	ENTER	1.310	1,500	0	1,965	0	12,428,783,834,700,880,279			B					140
15	AAC	20090701	00:00:00.000	ENTER	1.610	10,416	0	16,769.76	0		2,443,021,378,870,614,725		A					606
16	AAC	20090701	00:00:00.000	ENTER	1.260	800	0	1,008	0	3,980,296,868,089,774,903			B				140	
17	AAC	20090701	00:00:00.000	ENTER	1.570	10,640	0	16,704.8	0		3,019,473,372,611,958,729		A					606
18	AAC	20090701	00:00:00.000	ENTER	1.085	461	0	500.185	0	9,617,392,236,442,234,685			B				606	
19	AAC	20090701	00:00:00.000	ENTER	1.845	1,000	0	1,845	0		15,087,888,697,205,409,711		A					140
20	AAC	20090701	00:00:00.000	ENTER	1.320	5,000	0	6,600	0	14,165,957,609,012,303,084			B				606	
21	AAC	20090701	00:00:00.000	ENTER	1.300	2,000	0	2,600	0	11,563,661,767,927,919,676			B				140	

C. Implied Cost of Capital Estimation

We estimate the implied cost of equity capital (ICC) using two models: Ohlson and Juettner-Nauroth (2005) as implemented by Gode and Mohanram (2003), and Gordon and Gordon (1997). To generate the required earnings forecasts, we follow the cross-sectional approach of Hou, van Dijk, and Zhang (2012), which replaces analyst consensus forecasts with model-based expectations and substantially expands sample coverage. For each country and year, we estimate a pooled cross-sectional regression of future earnings on current firm characteristics (total assets, dividends, a dividend payer dummy, earnings, a negative earnings dummy, and accruals) using a rolling 10-year window. We estimate the model separately for each country to avoid mixing different currency units. The estimated coefficients are then applied to current-year characteristics to generate out-of-sample earnings forecasts $\hat{E}_{i,t+\tau}$ for $\tau = 1, \dots, 5$ years ahead.

We compute the ICC at the end of June of year t , using financial statements with fiscal year-ends from April of year $t-1$ through March of year t to ensure a minimum three-month reporting lag. Let P_t denote the stock price, $\hat{e}_{t+\tau}$ the per-share forecasted earnings, and $\hat{d}_{t+1} = \hat{e}_{t+1} \times k$ the expected per-share dividend, where k is the payout ratio.

Model 1: OJ (Ohlson and Juettner-Nauroth (2005); Gode and Mohanram (2003)). The Ohlson and Juettner-Nauroth (2005) model, as implemented by Gode and Mohanram (2003), prices the firm based on expected near-term abnormal earnings growth:

$$r_{OJ} = A + \sqrt{A^2 + \frac{\hat{e}_{t+1}}{P_t}(\hat{g}_{st} - (\gamma - 1))} \quad (15)$$

where $A = \frac{1}{2}[(\gamma - 1) + \hat{d}_{t+1}/P_t]$, $\gamma - 1 = \max(r_f - 0.03, 0)$ is the perpetual earnings growth rate, and $\hat{g}_{st}$ is the short-term earnings growth rate:

$$\hat{g}_{st} = \frac{1}{2} \left[\frac{\hat{e}_{t+3} - \hat{e}_{t+2}}{\hat{e}_{t+2}} + \frac{\hat{e}_{t+5} - \hat{e}_{t+4}}{\hat{e}_{t+4}} \right]$$

Model 2: Gordon (Gordon and Gordon (1997)). The Gordon and Gordon (1997) finite-horizon expected return model simplifies to a pure earnings yield:

$$r_{Gordon} = \frac{\hat{e}_{t+1}}{P_t} \quad (16)$$

This assumes that all growth beyond the forecast horizon is value-neutral (i.e., incremental investments earn exactly the cost of capital).

D. Price Delay and Return Comovement

Price delay. We measure the speed of price adjustment using the price delay measure of Hou and Moskowitz (2005). For each firm i in year t , we estimate the following two regressions using weekly returns:

$$r_{i,w} = \alpha_i + \beta_i r_{m,w} + \varepsilon_{i,w} \quad (17)$$

$$r_{i,w} = \alpha_i + \beta_{i,0} r_{m,w} + \sum_{k=1}^4 \beta_{i,k} r_{m,w-k} + \varepsilon_{i,w} \quad (18)$$

where $r_{i,w}$ is firm i 's return in week w and $r_{m,w}$ is the market return, defined as the equal-weighted average return of all other stocks in the same country excluding firm i itself. Price delay is then defined as:

$$D_i = 1 - \frac{R_{\text{restricted}}^2}{R_{\text{unrestricted}}^2} \quad (19)$$

where $R_{\text{restricted}}^2$ and $R_{\text{unrestricted}}^2$ are from equations (17) and (18), respectively. Higher values of D_i indicate slower incorporation of market-wide information into the stock price. We require a minimum of 26 weekly observations per firm-year.

Return comovement. We measure return comovement using the R^2 from a market model regression. For each firm i in year t , we estimate:

$$r_{i,w} = \alpha_i + \beta_{i,0} r_{m,w} + \beta_{i,1} r_{m,w-1} + \beta_{i,2} r_{m,w+1} + \varepsilon_{i,w} \quad (20)$$

where $r_{m,w}$ is the equal-weighted market return (excluding firm i), and $r_{m,w-1}$ and $r_{m,w+1}$ are the one-week lagged and leading market returns, respectively. The inclusion of lead and lag terms follows Morck, Yeung, and Yu (2000) and accounts for nonsynchronous trading. The R^2 from this regression captures the fraction of a stock's return variation explained by market-wide factors. Higher R^2 indicates greater return comovement and less firm-specific information in prices. We require a minimum of 30 weekly observations per firm-year.

E. The List of Co-location Events

The List of Co-location Events

This table lists the ten exchange-level adoption dates used in the co-location analysis. We begin with candidate dates reported in [Boehmer, Fong, and Wu \(2021\)](#) and [Aitken, Cumming, and Zhan \(2023\)](#), reconcile differences using exchange press releases and contemporaneous news reports, and exclude events with unavailable dates or unresolved dating ambiguity.

Stock Exchange	Country	Date
Toronto Stock Exchange (TSX)	CAN	Apr 2008
BSE Limited	IND	Feb 2010
National Stock Exchange of India (NSE)	IND	Jan 2010
SIX Swiss Exchange	CHE	Apr 2012
The Stock Exchange of Hong Kong (SEHK)	HKG	Oct 2012
London Stock Exchange	GBR	Sep 2009
Oslo Børs	NOR	Apr 2010
Singapore Exchange (SGX)	SGP	Jul 2011
Taiwan Stock Exchange (TWSE)	TWN	Oct 2010
Tokyo Stock Exchange (TSE)	JPN	Jun 2009

F. Supplementary Analysis

Table F1
ITCH DiD Results with Restricted Event Windows

This table reports ASX-ITCH difference-in-differences (DiD) estimates using the baseline specification but restricting the sample to observations within symmetric event windows around the ASX-ITCH introduction. Columns (1)–(3) use a $[-5, +5]$ window and Columns (4)–(6) use a $[-10, +10]$ window, where year 0 (2012) is excluded. The dependent variables are investment (Capex and Δ PPE) and external equity financing (Equity Issues), as indicated in the column headings. All variables are defined in Online Appendix Section A. All regressions include firm and year fixed effects and Size Group $\times$ SIC2 fixed effects. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	[-5, 5] Window			[-10, 10] Window		
	(1) Capex	(2) Δ PPE	(3) Equity Issues	(4) Capex	(5) Δ PPE	(6) Equity Issues
Treated $\times$ Post	-0.716 (-0.774)	-2.503** (-2.330)	-7.366** (-2.669)	-1.404* (-1.817)	-2.841** (-2.695)	-6.731** (-2.787)
Size	-3.564*** (-4.412)	-4.946*** (-3.757)	-10.126** (-3.076)	-3.212*** (-5.084)	-3.449*** (-3.912)	-8.060*** (-4.685)
Cash Flow	0.074 (1.440)	0.087 (1.375)	-0.132 (-0.774)	0.123** (2.378)	0.177** (2.335)	0.150 (0.846)
Past Profitability	-0.057 (-1.102)	0.022 (0.200)	-0.398** (-2.272)	-0.083 (-1.460)	-0.087 (-1.115)	-0.454** (-2.307)
Leverage			0.114 (1.336)			0.047 (0.984)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2	Yes	Yes	Yes	Yes	Yes	Yes
N	1,012	1,018	1,015	1,742	1,752	1,745
Adj. R ²	0.605	0.324	0.374	0.570	0.269	0.318

Table F2
Investment-to-Q Sensitivity: ASX-ITCH IPW and Co-location Evidence

This table reports investment- q sensitivity regressions for ASX-ITCH and exchange co-location. Q is lagged Tobin's q . Columns (1)–(2) report the existing inverse-probability-weighted (IPW) ASX-ITCH specifications. Columns (3)–(4) report staggered DiD estimates for the ten included co-location events using the $[-10, +10]$ event window and excluding the adoption year. The key interaction is $Treated \times Post \times Q$ for ASX-ITCH and $Co-location \times Q$ for co-location. Co-location regressions include firm and year fixed effects, Country $\times$ SIC2 fixed effects, and the controls reported in the table. t -statistics based on standard errors clustered at the firm level are in parentheses below each coefficient. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

	ASX-ITCH IPW		Co-location	
	(1) Capex	(2) Δ PPE	(3) Capex	(4) Δ PPE
Treated $\times$ Post $\times$ Q	0.613 (1.476)	0.702 (1.302)		
Co-location $\times$ Q			-0.133*** (-3.754)	-0.063 (-1.585)
Treated $\times$ Post	-2.543 (-1.593)	-3.789** (-2.163)		
Co-location			-0.094 (-0.725)	-0.585*** (-3.497)
Treated $\times$ Q	-0.342 (-0.753)	-0.010 (-0.014)		
Post $\times$ Q	-0.725** (-2.097)	-0.685* (-1.817)		
Q	1.018** (2.351)	0.971* (1.907)	0.560*** (16.035)	0.606*** (15.595)
Cash Flow $\times$ Treated $\times$ Post	-0.034 (-0.673)	-0.132 (-1.502)		
Cash Flow $\times$ Co-location			-0.012*** (-2.675)	-0.013** (-2.522)
Cash Flow	0.056 (1.415)	0.175** (2.344)	0.036*** (7.928)	0.064*** (10.595)
Size	-2.154*** (-3.498)	-1.450* (-1.866)	-1.042*** (-23.085)	-1.344*** (-26.185)
Past Profitability	0.005 (0.084)	-0.012 (-0.131)	0.079*** (18.363)	0.092*** (17.305)
GDP Growth			0.013 (0.913)	0.050*** (2.807)
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Size Group $\times$ SIC2 FE	Yes	Yes		
Country $\times$ SIC2 FE			Yes	Yes
N	1,725	1,733	113,968	119,334
Adj. R ²	0.609	0.285	0.474	0.215

Figure F3. Dynamic DiD Estimates from the Co-location Setting

This figure plots stacked inverse-probability-weighted dynamic difference-in-differences estimates around exchange co-location adoption for Capex, Δ PPE, Equity Issues, and the implied cost of equity measured using r_{OJ} . For each adoption cohort, the stack includes firms treated in that cohort and firms from later cohorts only while those firms remain untreated. The propensity-score model, weights, controls, and fixed effects follow Panel B of Table 8. Event time $t = -1$ is the omitted reference period. The adoption year, $t = 0$, is estimated rather than omitted. Standard errors are clustered at the firm level. The light vertical intervals report 90% confidence intervals, and the vertical dashed line marks adoption at $t = 0$.

